Statistical Guarantees for Sequential Decision Agents

Risk, Representation, and Verification

From Wald's decision theory and Markov decision processes
to evaluation, inference, and verifiable guarantees for agents

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An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

— Bellman (1957)

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Preface

This monograph is about one discipline: a claim about a sequential decision agent is only as strong as the chain that links it to data. The chain has three links. A **representation** link fixes what the decision problem is — who decides, from what information, under which family of models. An **estimation** link fixes what the data allow us to learn about a policy and with what precision. A **verification** link fixes which checks confirm that the first two links hold in a concrete system.

The book grows out of a doctoral course taught in Spanish (**Teoría Avanzada de Inferencia Estadística aplicada a la decisión secuencial y los agentes inteligentes**). The course book is broader; this volume is sharper. It keeps the representation theorem that anchors the course — a faithful one-step embedding of Wald decision problems into Markov decision processes — and builds outward in three directions the course only sketched: the structural frontier of that embedding, inference that stays valid while data still accumulate, and a calculus for deciding when a guarantee may be reported.

Conventions

Every numbered environment carries its hypotheses; every proof is in Appendix A or is explicitly marked as a sketch with a pointer to its source. Every displayed number is computed by a script shipped with the book (Appendix D), with fixed seeds. An OPEN box declares a gap instead of filling it with plausible prose; several remain, and they are meant to remain until someone closes them — possibly the reader.

The synthetic example thread SSCH (a hypertension support system) is used to make every calculation checkable; its figures are not clinical estimates, and the book says so each time it uses them. The single real-data study (Chapter 11) is the one place where the protocol is executed end to end on data that the authors did not generate.

Acknowledgments of sources

The estimation part draws on the off-policy evaluation literature, particularly the work of Kallus and Uehara and of Shi, Zhang, Lu and Song; the anytime-valid part on the program of Ramdas, Vovk and their collaborators; the decision-theoretic spine on Wald, Ferguson, Berger, and Puterman. Each result used from these works is stated with its own hypotheses and cited at the point of use; the annotated bibliography conventions of the course book apply here in full.

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DRAFT v0.1 — Statistical Guarantees for Sequential Decision Agents

Introduction

An imperfect signal can be enough to change a decision. A monitoring system observes a measurement and chooses between routine attention and intensified care. To choose, it needs more than a prediction: it must weigh the cost of each error and the information that was available when it acted. If the intervention changes next period's measurements, the same question acquires a temporal dimension.

Statistical decision theory and Markov decision processes (MDPs) describe those two situations. Wald compares rules by the loss they produce under each plausible model. Dynamic programming organizes present and future consequences of actions. This book studies the relation between the two languages — what can be preserved when passing from one to the other, and what demands a new argument — and then asks the harder question: **what can be learned about a decision rule from data, and with what guarantees?**

The thesis, stated narrowly

A risk-preserving representation exists: any finite Wald decision problem can be written as a family of one-decision MDPs, indexed by the unknown parameter, in such a way that the return of the copied policy equals minus the risk of the rule (Chapter 4). Bayes, minimax, and admissibility travel through this identity — within a fixed class of rules. What does **not** travel automatically is everything inferential: an identity between objectives produces no data, no estimator, no interval. The traceability discipline of this book consists in saying, at each step, which of the two kinds of statement is being made.

Definition 0.1 (Informal statement of the representation). A rule that maps an observation to an action can be read as a policy on that same observation. If the reward in each model is the negative loss, the average return of the policy coincides with minus the risk of the rule. The parameter remains unknown and is not part of the information available for acting. Theorem 4.3 gives the conditions and the exact identity; its converse frontier is Theorem 4.9.

DRAFT version — Statistical Decision Theory for Sequential Problems

Object	Role in the representation	Distinction to keep
Θ	Unknown index of the model family	The decision maker does not observe the parameter
$\mathcal{X}$	Observation received by the policy	The available information is the same
$\mathcal{A}$	Common action set	The same class of rules is compared
$L(\theta, a)$	Negated reward	Evaluation keeps its dependence on the model
$R(\theta, \delta)$	Average negative return	The average uses the distribution of the observation
Interval, test, estimator	Statistical procedure on data	Its guarantee needs additional hypotheses

Table 0.1: Objects that must be preserved across the representation. Inferential guarantees are justified separately, for the actual experiment.

Reading map

Chapters 1–3 fix the two frameworks: experiments and sufficiency, Wald, finite discounted MDPs. Chapter 4 proves the representation and its structural frontier; Chapter 5 the Bayesian stopping extension. Chapters 6–9 are the estimation part: identification and its failures, DM/IS/DR with their variance analysis, anytime-valid policy evaluation (the monograph's main addition to the course material), and learned policies. Chapters 10–13 close: transfer certificates, the real-data study, the language-agent frontier, and open problems. Appendix A carries every proof whose statement appears in the body and is not explicitly credited to a source.

Prerequisites

Conditional expectation, convergence in distribution, measurable structure; estimation, testing, and confidence intervals at the graduate level; norms, projections, and contraction arguments. Appendix B is a reference summary of every invoked tool, organized by where the book invokes it.

OPEN — Exercises are not yet ported to this draft. The companion course book contains a full set per chapter; they will be adapted once the English text is stable.

DRAFT v0.1-M — Statistical Guarantees for Sequential Decision Making

Part I

Representation: what a decision problem is

Statistical experiments organize what can be learned from data; Wald's framework compares rules by risk; dynamic programming adds the future. One representation theorem and one stopping result tie these languages together with information constraints kept explicit.

Chapter 1

Statistical experiments and comparison

Before deciding, one must say what can be learned. A statistical experiment precisely specifies the data and their possible distributions. This chapter fixes that vocabulary — experiments, statistics, sufficiency, completeness — and the comparison of experiments that will later tell us what the representation of Chapter 4 preserves. The classic results are stated with their hypotheses; proofs are in Appendix A.

Experiments and statistics

Definition 1.1 (Statistical experiment). An experiment is a triple $\mathcal{E} = \left(\mathcal{X}, \mathcal{F}, (P_\theta)_{\theta\in\Theta}\right)$: a measurable space of observations and a family of probability measures indexed by a parameter set. The family is part of the object: it declares what the observation can reveal about what is unknown. A statistic is a measurable map $T : \mathcal{X} \to \mathcal{T}$. An experiment is dominated when all P_θ share a density with respect to one sigma-finite measure.

Data reduction: sufficiency

Definition 1.2 (Sufficiency). T is sufficient for θ when a version of the conditional distribution of the data given T can be chosen not to depend on θ. Sufficiency is a property of the experiment, not of a storage format: declaring a field in a program does not create it (cf. Note 1.9).

Theorem 1.3 (Neyman–Fisher factorization). *In a dominated experiment with standard Borel spaces, T is sufficient if and only if the densities factor as $f_\theta(x) = g_\theta(T(x))h(x)$ for measurable g_θ, h; h may be chosen independent of θ.*

Proof. In one direction, a common conditional kernel and the Halmos–Savage dominating mixture give $\mathrm{d}P_\theta/\,\mathrm{d}Q = g_\theta(T)$; in the other, the conditional expectation under the mixture is seen not to depend on θ. Full argument and regularity conditions: Section A.1.1. □

Definition 1.4 (Minimal sufficiency). T is minimal sufficient if it is sufficient and a function of every sufficient statistic (up to null sets). The likelihood-ratio classes criterion: under the precise measurability hypotheses of Appendix A, the statistic that identifies the proportionality classes of the likelihoods is minimal sufficient.

Definition 1.5 (Completeness). T is complete if $\mathbb{E}_\theta[g(T)] = 0$ for every θ implies $g(T) = 0$ almost surely under every P_θ. Completeness is what turns an identity in expectation into an identity of functions.

Theorem 1.6 (Basu). *If T is boundedly complete and sufficient, and A is ancillary (its distribution does not depend on θ), then T and A are independent under every P_θ. Proof: Section A.1.2.*

Completeness holds in exponential families whose natural parameter space contains an open set (Appendix B). Trajectory counts of a Markov chain do **not** inherit the completeness of a fixed-size

multinomial experiment: their random totals define another experiment — this warning will matter when agents store counts of transitions (Chapter 6).

An example of what risk comparison buys: James–Stein

Example 1.7 (James–Stein). For $X \sim N_p(\theta, I_p)$, $p \geq 3$, squared-loss risk, the rule $\delta_0(X) = X$ has constant risk p, while

$$\delta_{\mathrm{JS}}(X) = \left(1 - \frac{p-2}{\|X\|^2}\right) X$$

has risk

$$p - (p-2)^2 \mathbb{E}_\theta\left[\frac{1}{\|X\|^2}\right] < p.$$

The expectation is finite for $p \geq 3$; the identity is Stein's lemma applied to $\div\,(x/\|x\|^2) = (p-2)/\|x\|^2$ (Section A.1.3). The dimension at stake is that of the parameter, not the number of observations of a scalar mean.

Comparing experiments

Note 1.8 (Blackwell informativeness, at reading level). An experiment $\mathcal{E}$ is more informative than $\mathcal{E}'$ if every risk function achievable under $\mathcal{E}'$ is achievable under $\mathcal{E}$ (equivalently: $\mathcal{E}'$ is a garbling of $\mathcal{E}$, or their Le Cam deficiencies vanish in one direction). Sufficiency is the special case where a statistic loses nothing. This book uses the vocabulary only where it sharpens a claim: Chapter 4's frontier and Chapter 10's certificates. The systematic theory is in Torgersen (1991) and Blackwell (1951).

Note 1.9 (Three distinct requirements on a "state"). A typed data structure says what is stored. Statistical sufficiency says which sample information about a parameter is retained. An information state for control must allow computing the relevant future distributions and costs. Markovianity is a property of a transition law. None of these follows from declaring a `StateSchema`. Chapter 6 gives the failure taxonomy; Chapter 4 and Chapter 5 give two settings where the right object is provably enough.

DRAFT v0.1-M — Statistical Foundations for Sequential Decision Making

Chapter 2

Wald's decision theory

An estimate summarizes what the data allow one to know; a decision adds the consequences of acting. Wald's framework reunites observations, models, actions, and losses, and compares complete rules under uncertainty. This chapter develops a single decision from one sample; the MDP chapter adds future consequences, and Chapter 4 makes their relation exact.

Problems, rules, and risk

Definition 2.1 (Statistical decision problem). A problem is specified as

$$\mathcal{E} = \left(\Theta, \mathcal{X}, (P_\theta)_{\theta\in\Theta}, \mathcal{A}, L\right),$$

with measurable structures everywhere and functions measurable as needed. Losses are nonnegative or integrable enough that the risks considered are finite; bounded losses of either sign are admissible.

Definition 2.2 (Rules and classes). A deterministic rule is a measurable $\delta : \mathcal{X} \to \mathcal{A}$; a randomized rule is a kernel $d(a|x)$ with $d(a|x) \geq 0$ and $\sum_a d(a|x) = 1$ (finite actions). Randomization uses available information plus randomness independent of the parameter. The class of allowed rules $\mathcal{D}$ must be declared; comparisons change when it changes.

Risk: for a deterministic rule, $R(\theta, \delta) = \mathbb{E}_\theta[L(\theta, \delta(X))]$; for randomized, $R(\theta, d) = \mathbb{E}_\theta\left[\sum_a d(a|X)L(\theta, a)\right]$. Risk is a function of θ; in general no single rule minimizes all its components.

Definition 2.3 (Dominance and admissibility). δ_1 dominates δ_2 in $\mathcal{D}$ if $R(\theta, \delta_1) \leq R(\theta, \delta_2)$ for all θ with strict inequality somewhere. A rule is admissible when nothing in its class dominates it. Admissibility discards uniform improvements; it does not order rules nor establish optimality for a particular prior.

Criteria: Bayes, minimax, and their ties

Definition 2.4 (Prior and Bayes risk). A prior ν defines $\rho_\nu(\delta) = \int R(\theta, \delta)\nu(\mathrm{d}\theta)$ and the optimal Bayes value $b(\nu) = \inf_{\delta\in\mathcal{D}} \rho_\nu(\delta)$. A rule attaining the infimum is a Bayes rule — optimal relative to the prior, loss, and declared class.

Theorem 2.5 (Posterior loss minimization). *Suppose a regular posterior $\nu(\mathrm{d}\theta|x)$ exists, $\mathcal{A}$ is finite, and L is bounded and measurable. Choose, for each x, an action minimizing the posterior loss $\ell_\nu(x, a) = \int L(\theta, a)\nu(\mathrm{d}\theta|x)$, with a fixed tie-break. The resulting rule is measurable and Bayes among all rules, randomized included; every Bayes rule concentrates on minimizers for predictively almost every observation. Details: Section A.2.1.*

Theorem 2.6 (Minimax certified by Bayes). *If δ_ν is Bayes and $\sup_\theta R(\theta, \delta_\nu) = \rho_\nu(\delta_\nu) < \infty$ — in particular if its risk is constant on all of Θ — then it is minimax. Constancy only on the prior's support does not suffice (Section A.2.2).*

Theorem 2.7 (Essentially complete class, finite problem). *With finite $\Theta, \mathcal{X}, \mathcal{A}$, finite loss, and all randomized rules allowed, every admissible rule is Bayes for some prior (possibly assigning zero mass to some parameters), and every rule has a Bayes rule whose risk is nowhere larger. Convexity and a supporting hyperplane; proof in Section A.2.3. This is a statement about the finite problem, not a general Bayes-limits theorem.*

Proposition 2.8 (Bayes rules are admissible under a positive prior). *If Θ is finite, $\nu(\theta) > 0$ everywhere, and δ_ν has finite Bayes risk, then δ_ν is admissible within the same class.*

Tests as decision rules and the thread example

Proposition 2.9 (Tests as rules). *Hypotheses partition Θ; accepting/rejecting have losses $w_I, w_{II} > 0$ for the two error types. The Bayes rule rejects when $w_I \mathbb{P}(\Theta_0|X) < w_{II}\mathbb{P}(\Theta_1|X)$. Frequentist level control is an additional constraint; choosing losses and a prior does not guarantee a prescribed level.*

Example 2.10 *(synthetic)* **(SSCH: signal, loss, and class of rules).** $\theta \in \{0, 1\}$: need for intensified follow-up. Signal $X \in \{0, 1\}$ with $P_0(X = 1) = 0.2$, $P_1(X = 1) = 0.8$. Losses 1 for intensifying without need, 4 for missing the need. With prior $\nu(1) = 0.3$:

Rule	$R(0)$	$R(1)$	**Bayes risk**
Always routine	0	4	1.20
Always intensified	1	0	0.70
Follow the signal	0.20	0.80	0.38
Invert the signal	0.80	3.20	1.52

Following the signal is Bayes and minimax among the four deterministic rules. Allowing randomization, a better worst-case rule intensifies always if $X = 1$ and with probability $\frac{3}{8}$ if $X = 0$: both risks equal $\frac{1}{2}$, and the prior $\nu^*(1) = \frac{1}{2}$ certifies minimaxity via Theorem 2.6 (verification: `verificacion/m_ch2_ssch.py`).

Note 2.11 (What each criterion does not promise). Minimax controls the worst expected risk inside the declared family; it is not safety per trajectory and the least worst case may still be bad. Posterior minimality is relative to the prior. Switching criteria using the same data used for evaluation defines a new procedure whose risk must be analyzed as a whole.

Readings

Wald (1950) for the foundation; Berger (1985) and Ferguson (1967) for losses, Bayes and minimax rules; each read with its class of rules and hypotheses.

Chapter 3

Markov decision processes

Chapter 2 separated data, rules, and consequences. When an action modifies future opportunities, comparing only immediate losses is insufficient. A Markov decision process describes those consequences and computes policies that account for them. The statistical problem returns when rewards or transitions must be learned from data — Chapters 6–9.

This chapter develops planning guarantees in finite, discounted, known MDPs, and keeps them distinct from the guarantees needed to learn the model.

The model and its values

Definition 3.1 (MDP). $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ with finite nonempty states and actions, $P(s'|s,a) \geq 0$, $\sum_{s'} P(s'|s,a) = 1$, $|r(s,a)| \leq R_{\max} < \infty$, $0 \leq \gamma < 1$. An initial distribution ρ defines the aggregate value $J_\rho^\pi = \sum_s \rho(s) V^\pi(s)$. Random rewards with conditional mean $r(s,a)$ are allowed; integrability and the corresponding bound suffice for the value formulas.

Definition 3.2 (Policies and Markov property). A stationary randomized policy assigns $\pi(a|s)$; deterministic assigns $\pi(s)$. In the general class a policy may depend on time and on the whole observed history. Policies use only information available at the time of acting. The Markov property is

$$\mathbb{P}(S_{t+1} = s' | S_0, A_0, \ldots, S_t, A_t) = P(s'|S_t, A_t),$$

plus a reward conditional mean determined by (S_t, A_t) — both tied to the state actually chosen for the model.

Definition 3.3 (Value functions). For a stationary policy,

$$V^\pi(s) = \mathbb{E}_s^\pi\left[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t)\right], \quad Q^\pi(s,a) = r(s,a) + \gamma \sum_{s'} P(s'|s,a) V^\pi(s'),$$

the latter also well-defined when $\pi(a|s) = 0$ (no conditioning on an impossible event). Elementary bound: $|V^\pi(s)| \leq R_{\max}/(1-\gamma)$.

Bellman equations, contraction, and computation

Theorem 3.4 (Bellman expectation equation).

$$V^\pi(s) = \sum_a \pi(a|s)\left[r(s,a) + \gamma \sum_{s'} P(s'|s,a) V^\pi(s')\right].$$

Proof. Split the return, condition on first action and next state, use stationarity; the geometric bound justifies exchanging expectation and series. Full details: Section A.3.1. □

Definition 3.5 (Bellman operators). For $V \in \mathbb{R}^{|\mathcal{S}|}$,

$$(T^\pi V)(s) = \sum_a \pi(a|s) \left[r(s,a) + \gamma \sum_{s'} P(s'|s,a) V(s') \right],$$

$$(T^* V)(s) = \max_{a \left[r(s,a) + \gamma \sum_{s'} P(s'|s,a) V(s') \right]}.$$

Theorem 3.6 (Sup-norm contraction). *For $T = T^\pi$ or $T = T^*$,*

$$\|TV - TW\|_\infty \le \gamma \|V - W\|_\infty.$$

Proof. The difference of two maxima is bounded by the maximum absolute difference; each averaging is bounded by $\|V - W\|_\infty$; Section A.3.2. □

Corollary 3.7 (Unique fixed points). *Banach's theorem on the finite space with the sup norm gives unique fixed points: V^π of T^π and V^* of T^*.*

Theorem 3.8 (Optimality). *In the finite discounted MDP there exists a deterministic stationary optimal policy, and $V^*(s) = \max_{a \left[r(s,a) + \gamma \sum_{s'} P(s'|s,a) V^*(s') \right]}$. The optimal value is unique; optimal policies may not be. A greedy policy at the fixed point attains the optimum, including against history-dependent policies; Section A.3.3.*

Theorem 3.9 (Value iteration). $\|V_k - V^*\|_\infty \le \gamma^k \|V_0 - V^*\|_\infty$; *with* $V_0 = 0$, $\gamma^k R_{\max}/(1-\gamma)$.

Corollary 3.10 (Stopping certificates). *Let $e = \|T^* V - V\|_\infty$ and π_V greedy at V. Then*

$$\|V - V^*\|_\infty \le \frac{e}{1-\gamma}, \quad \|V^* - V^{\pi_V}\|_\infty \le \frac{2\gamma e}{(1-\gamma)^2}.$$

Two distinct precision goals: a value error and a policy loss have different residual thresholds. Proofs: Section A.3.5.

Theorem 3.11 (Policy improvement and finite termination). *A policy greedy at V^π satisfies $V^{\pi'} \ge V^\pi$; with the tie-keeping rule, an actual change is a strict improvement somewhere, so policy iteration visits no policy twice and stops at an optimal one. It is a monotonicity-and-finiteness argument, not a contraction; Section A.3.4.*

Bandits: a pointer with two computed lessons

Theorem 3.12 (UCB1 under bounded rewards). *For rewards in $[0,1]$ and horizon $T \ge K$, UCB1 with*

$$\hat{\mu}_a + \sqrt{\frac{2 \log t}{N_a}}$$

has pseudo-regret

$$\mathcal{R}_T \le 8 \sum_{a:\Delta_a > 0} \frac{\log T}{\Delta_a} + \left(1 + \frac{\pi^2}{3}\right) \sum_a \Delta_a,$$

Auer et al. (2002). Gap-dependent bound; the asymptotic constant is not claimed optimal (Lai–Robbins-type limits are quoted, with their hypotheses, in Appendix B).

Example 3.13 *(synthetic)* **(Posterior sampling is not posterior optimality).** Two arms with independent posteriors $\tilde{\mu}_1 \sim \mathrm{Beta}(4,1)$ and $\tilde{\mu}_2 \sim \mathrm{Beta}(1,1)$ (means $0.8, 0.5$). With one round left,

the Bayes action picks arm 1 (predictive 0.8). Thompson sampling picks arm 1 with $\mathbb{P}(\tilde{\mu}_1 > \tilde{\mu}_2) = \mathbb{E}[\tilde{\mu}_1] = 0.8$ (the second arm's prior is uniform) and collects $0.8 \cdot 0.8 + 0.2 \cdot 0.5 = 0.74$. Sampling from the posterior and acting on the sample is not minimizing posterior risk; the difference is 0.06 here (`verificacion/m_ch3_thompson.py`).

Example 3.14 *(synthetic)* **(SSCH reduced two-state model).** States g, b, actions 0,1, $\gamma = 0.9$, deterministic transitions: (g,0) gives 1 and goes to b; (g,1) gives 0.5, stays; (b,0) gives 0, stays; (b,1) gives -0.5, goes to g. $V^* = (5, 4)$ with action 1 in both states: at g, $\max\{1 + 0.9 \cdot 4,, 0.5 + 0.9 \cdot 5\} = 5$; at b, $\max\{0.9 \cdot 4,, -0.5 + 0.9 \cdot 5\} = 4$. Iterates from zero: $(1, 0), (1.4, 0.4), (1.76, 0.76)$. Maximizing immediate rewards would choose 0 in both states — the example isolates why future consequences can pay a present cost (`verificacion/m_ch3_bellman.py`).

Note 3.15 (Known model versus learned model). A contraction certifies planning inside the model used; it certifies nothing about the environment's true value if the model was estimated. Estimating transition tables is a finite-dimensional problem in a fixed finite space — "tabular" does not mean nonparametric — and in continuous state spaces it is a nonparametric one. A scalar objective J_ρ^π inside an infinite-dimensional family is the semiparametric problem of Part II.

DRAFT Version — Statistical Guarantees for Sequential Decision Agents

Chapter 4

The risk-preserving representation

An agent receives information and chooses an action whose consequences depend on an unknown environment. Decision theory describes that choice through a loss and its expected risk; MDPs describe actions, rewards, and transitions. Connecting the languages requires preserving the information available for deciding and the criterion used for comparison.

This chapter constructs an exact representation of a statistical decision problem as a **family of one-decision MDPs**, indexed by the unknown parameter, and then proves the structural frontier of that construction: exactly when a sequential problem admits no such representation with its full strategy class. The scope is a representation that preserves risk — not a category equivalence, and no composition of decision rules is ever used.

The two frames, recalled

Definition 4.1 (Statistical decision problem). Let Θ be finite nonempty, $\mathcal{X}$ finite or countable, $\mathcal{A}$ finite nonempty, and $L : \Theta \times \mathcal{A} \to \mathbb{R}$ bounded. A problem is $\mathcal{D} = (\Theta, \mathcal{X}, (P_\theta), \mathcal{A}, L)$; a deterministic rule $\delta : \mathcal{X} \to \mathcal{A}$ is chosen without knowing θ; its risk is

$$R(\theta, \delta) = \sum_{x \in \mathcal{X}} P_\theta(x) L(\theta, \delta(x)).$$

All comparisons happen within the class $\mathcal{D}_0$ of deterministic rules, unless stated otherwise.

Definition 4.2 (One-decision MDP with terminal state). An initial state S_0 in $\mathcal{X}$ is observed, an action $A_0 \in \mathcal{A}$ is chosen, one reward is received, and the process reaches a terminal state s_{fin} of zero further reward. A deterministic policy is a map $\pi : \mathcal{X} \to \mathcal{A}$. For the model indexed by θ, distinguish

$$V_\theta^\pi(x) = r_\theta(x, \pi(x)), \quad J_\theta(\pi) = \sum_x P_\theta(x) V_\theta^\pi(x) :$$

value conditioned on the initial state, and return averaged over the initial distribution. Discount one is admissible: a single reward, finite horizon.

The representation theorem

Theorem 4.3 (Risk-preserving representation). *For the problem of Definition 4.1, construct for each θ an MDP $\mathcal{M}_\theta$ with initial distribution P_θ, reward $r_\theta(x, a) = -L(\theta, a)$, and deterministic transition to the terminal state. The correspondence $F(\delta) = \pi_\delta$, $\pi_\delta(x) = \delta(x)$, is a bijection between deterministic rules and deterministic policies on the decision states of this family. For every θ and every rule,*

$$J_\theta(\pi_\delta) = -R(\theta, \delta).$$

One policy is used under all models. The bijection refers to pointwise functions; two distinct rules may share the same risk, for instance if they differ only on observations that are null under every model.

Proof. Rule and policy have the same domain and codomain, and copying them in either direction yields inverse maps. By the reward definition,

$$J_\theta(\pi_\delta) = \sum_x P_\theta(x)[-L(\theta, \delta(x))] = -R(\theta, \delta).$$

Boundedness gives absolute convergence; the terminal transition adds no future rewards. More in Section A.4.1. The three core identities of this chapter are the object of the machine-checked verification pilot (`VERIFICACION_LEAN_PLAN.md`, Phase 0). □

Note 4.4 (Scope of the construction). The initial state is observable; the index θ is not. The family evaluates a common policy; it does not hand the agent the true model, and the evaluative reward need not be observable after acting — a statistical loss may depend on a truth that never becomes visible, so this is not, by itself, a reinforcement-learning reward protocol. The identity uses J_θ, not $V_\theta^\pi(x)$; it does not equate a general MDP with a one-decision statistical problem.

What travels: criteria within a fixed class

Corollary 4.5 (Bayes criterion). *For a prior ν, with $\rho_\nu(\delta) = \sum_\theta \nu(\theta) R(\theta, \delta)$ and $J_\nu(\pi) = \sum_\theta \nu(\theta) J_\theta(\pi)$, one has $J_\nu(\pi_\delta) = -\rho_\nu(\delta)$: minimizers of Bayes risk correspond to maximizers of average return. The joint law used is $\nu(\theta) P_\theta(x)$; it cannot generally be replaced by an averaged reward without conditioning on x.*

Corollary 4.6 (Minimax criterion).

$$\inf_{\delta \in \mathcal{D}_0} \sup_{\theta \in \Theta} R(\theta, \delta) = -\sup_{\pi} \inf_{\theta \in \Theta} J_\theta(\pi).$$

When optima exist, minimax rules correspond to maximin policies. Randomizing both sides enlarges the classes consistently; a deterministic optimum may cease to be optimal in the enlarged class (cf. Example 2.10).

Theorem 4.7 (Admissibility as non-dominated return). *δ is admissible in $\mathcal{D}_0$ if and only if no policy of the corresponding class satisfies $J_\theta(\pi') \geq J_\theta(\pi_\delta)$ for all θ with strict inequality somewhere. The order compares the vector of returns indexed by **models**, not by observed initial states; the conclusion is a sign change in the definition of dominance (Section A.4.2).*

The frontier: when the law of the data depends on the rule

The construction above is a **folding** of one decision into one observation. Why does it break in honest sequential problems? Because in an MDP the law of the history depends on the policy, while a statistical experiment assigns one law per parameter. This can be made into a theorem, which is new in this monograph (its course-book companion asserts the difficulty in prose).

Definition 4.8 (Sequential problem with observation channel). Fix finite spaces and horizon H. A sequential problem family indexed by θ consists of an initial law $\mu_\theta(s)$, a per-time observation kernel $q_\theta(y_{t+1} | h_t, a_t)$, reward kernels, and an action set; a (history-dependent) strategy σ assigns actions $a_t = \sigma(h_t)$ with $h_t = (s_0, a_0, y_1, \ldots)$. The channel is **policy-independent at t** if $q_\theta(\cdot | h_t, a_t)$ does not depend on a_t for all reachable h_t.

Theorem 4.9 (Structural frontier of risk-preserving folding). *Suppose there exist θ, a time $t \geq 0$, a history h_t reached with positive probability under θ by the two prefix-agreeing strategies below, and actions $a \neq a'$ such that*

$$q_\theta(B|h_t, a) \neq q_\theta(B|h_t, a')$$

*for some event B on the next observation. Then there is no family $\left(\tilde{P}_\theta\right)$ of laws on the trajectory space and no loss $\tilde{L}$ such that, for **every** history-dependent strategy σ, the expected total reward of σ under θ equals minus the Wald risk of an associated rule under $\tilde{P}_\theta$ with the same averaging over trajectories. In words: when actions change the law of future observations, no Wald experiment folds the whole sequential problem with its strategy class.*

Proof. Assume such $\left(\tilde{P}_\theta, \tilde{L}\right)$ existed. Let σ, σ' agree strictly before t and take $\sigma(h_t) = a$, $\sigma'(h_t) = a'$. The law of the full trajectory induced by a strategy is then fixed; denote it $\mathbb{P}_\theta^\sigma$. A Wald experiment assigns one law $\tilde{P}_\theta$ to trajectories, so both induced laws would have to equal $\tilde{P}_\theta$ on the cylinder event $E = \{h_t \text{ occurs}, Y_{t+1} \in B\}$. But

$$\mathbb{P}_\theta^\sigma(E) = \mathbb{P}_\theta^\sigma(h_t) \cdot q_\theta(B|h_t, a)$$

and likewise for σ' with a'. By prefix agreement, $\mathbb{P}_\theta^\sigma(h_t) = \mathbb{P}_\theta^{\sigma'}(h_t) > 0$; hence $\mathbb{P}_\theta^\sigma(E) \neq \mathbb{P}_\theta^{\sigma'}(E)$, contradicting $\tilde{P}_\theta(E)$ being both. The argument is exact in finite spaces; in general measurable spaces the same cylinder computation applies with regular versions. Section A.4.3 discusses the boundary — unreachable histories exempted by the hypothesis — and the two positive special cases used in this book. □

Proposition 4.10 (The two admissible folds). *(i) Horizon one is always foldable: with no future observations, there is no law for actions to change (Theorem 4.3). (ii) Uncontrolled observe-and-stop problems are foldable at the level of information states: the per-observation channel $q_\theta(y)$ is policy-independent, and control acts only on continuation; Chapter 5 gives the exact dynamic program and its hypotheses. Beyond these cases, a risk-preserving representation with the full strategy class fails by Theorem 4.9.*

Note 4.11 (How to read this frontier). The theorem is not "the representation is limited"; it is the precise reason, and it doubles as a checkable condition: to fold a sequential system into one experiment, verify that actions do not enter the observation kernel within the episode. Learning-enabled agents usually violate it — their actions change what is observed next — which is exactly why Part II must treat estimation separately instead of transferring guarantees.

Chapter 5

Bayesian stopping and information states

The representation of Chapter 4 is one decision. Its first honest sequential extension is a Bayesian problem of observing and stopping: the parameter is fixed and unknown, observations arrive under a known, policy-independent channel, and the only control is whether to pay to see one more observation or to decide now. Within its hypotheses — finite spaces, explicit prior, finite horizon — the problem admits exact dynamic programming on the posterior, and the posterior with the remaining horizon is provably an **information state**: everything the history can contribute to an optimal continuation.

The model

A fixed unknown $\theta \in \Theta$ (finite), a prior b_0, observations in a finite $\mathcal{Y}$, conditionally independent given θ with known $q_\theta(y)$. At each instant: decide a terminal action, or pay $c \geq 0$ to observe once more; at most T observations; at the horizon one must decide.

With current posterior b, the predictive law and Bayes update are

$$m_b(y) = \sum_\theta b(\theta) q_\theta(y), \quad B(b,y)(\theta) = \frac{b(\theta) q_\theta(y)}{m_b}(y),$$

used when $m_b(y) > 0$ (arbitrary versions otherwise, with no effect on expectations). The expected loss of deciding a is $\ell(b,a) = \sum_\theta b(\theta) L(\theta, a)$.

Theorem 5.1 (Bayesian observe-and-stop dynamic program). *In the finite model above there is a Bayes-optimal strategy that depends on the history only through the number k of observations remaining and the posterior b. Its minimal remaining risk satisfies*

$$W_0(b) = \min_{a \in \mathcal{A}} \ell(b,a),$$

and for $k \geq 1$,

$$W_k(b) = \min\left\{ \min_{a \in \mathcal{A}} \ell(b,a),, c + \sum_{y \in \mathcal{Y}} m_b(y) W_{k-1}(B(b,y)) \right\}.$$

The strategy chooses an alternative attaining the minimum; ties may be broken by any fixed rule.

Proof, complete (deployed form promised by the course book). Write any strategy with k observations left as: current choice, then continuation strategies for each possible y. Two claims make the induction work in both directions.

(i) The value depends on history only through (k, b). Given the history, the law of the next observation is $m_b(y)$, a function of b alone, and the posterior after y is $B(b, y)$. The immediate costs are c or $\ell(b, a)$, also functions of b. Hence the continuation problem after any history inducing b is the same problem with parameters $(k-1, B(b, y))$; its optimal value is $W_{k-1}(B(b, y))$ once that value is defined independently of the history, which is precisely the induction hypothesis at $k-1$.

(ii) The recursion gives the optimal value and an attaining strategy. For $k = 0$ the only choice is to decide; the best terminal loss is $\min_a \ell(b, a)$. Suppose the claim for $k-1$ and consider k: any strategy either decides now — best value $\min_a \ell(b, a)$ — or observes, paying c and then facing $(k-1, B(b, y))$ with probability $m_b(y)$. By (i) and the induction hypothesis, the best observation-continuation has value $c + \sum_y m_b(y) W_{k-1}(B(b, y))$, attained by following the $(k-1)$-optimal strategy after each y. The better of the two alternatives is optimal, and choosing it realizes the minimum. Finiteness of $\Theta, \mathcal{Y}, \mathcal{A}$ makes every minimum attained and every selection measurable; versions at $m_b(y) = 0$ are immaterial since the corresponding terms vanish in every expectation. This closes the deferred proof of the course book; see also Section A.4.4 for the role of the information-state condition. □

Corollary 5.2 (Bellman form for Bayesian return). *With $U_k(b) = -W_k(b)$,*

$$U_0(b) = \max_a [-\ell(b, a)], \quad U_k(b) = \max\left\{ \max_a [-\ell(b, a)],, -c + \sum_y m_b(y) U_{k-1}(B(b, y)) \right\} :$$

reward $-c$ for observing, terminal expected reward $-\ell(b, a)$. The hidden parameter stays fixed; updating acts on beliefs over it. This is a statement of Bayesian optimality on information states; nothing here claims uniform frequentist optimality over θ.

Note 5.3 (Markovianity is not sufficiency). Let X_1, X_2 be i.i.d. Bernoulli(θ). The sequence is Markov, but the current state X_2 is not sufficient for θ in the sample (X_1, X_2): $\mathbb{P}_\theta(X_1 = 1 | X_2) = \theta$ depends on θ. The condition that matters for control is that, for a summary Z_t of the history, the law of (R_t, Z_{t+1}) given (H_t, A_t) factor through (Z_t, A_t); ordinary sufficiency for a static parameter does not give that. In Theorem 5.1, (k, b) satisfies it because the predictive law and update are computable from (k, b) alone (Section A.4.4).

Example 5.4 *(synthetic)* **(SSCH: whether to observe at all).** Continuing Example 2.10: deciding immediately with prior $\nu(1) = 0.3$ has minimal risk $\min\{1.2,, 0.7\} = 0.7$. One signal may be acquired at cost $c = 0.05$, then one decides. The posteriors are $12/19$ after $X = 1$ and $3/31$ after $X = 0$, with marginals $\mathbb{P}(X = 1) = 0.38$. The minimum posterior losses are $7/19$ and $12/31$, and

$$\mathbb{P}(X = 1) \cdot \frac{7}{19} + \mathbb{P}(X = 0) \cdot \frac{12}{31} = 0.14 + 0.24 = 0.38,$$

so observing costs $0.05 + 0.38 = 0.43 < 0.7$: observe. The gain comes from information that can change the action — an application of the recursion with one observation available, not an optimality claim for longitudinal clinical care (`verificacion/m_ch5_stopping.py`).

Note 5.5 (Where this stops). Controlled observations (q depending on actions) or latent states that evolve make the channel policy-dependent and fall outside Theorem 5.1 and outside folding, by Theorem 4.9: then one must specify the control kernel and the Bayesian filter explicitly, or accept a problem of another class.

Readings

Sequential analysis and Bayesian sequential decision: DeGroot (1970); Ferguson (1967) information states in stochastic control as in the notes of Tedrake (belief state and planning under uncertainty).

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Part II

Estimation with someone else's data

Evaluating a policy means identifying its functional in the observed law, estimating it under the actual sampling regime, and quantifying uncertainty at the right scale — including while the experiment still runs.

Chapter 6

Identification and observable contradictions

An identity between risk and return transfers a comparison criterion; it produces no new data and justifies no confidence interval by itself. The estimation part of the book starts by separating four questions that are independently answerable and often silently collapsed:

1. **Is the objective preserved?** (representation — Part I)
2. **Do the data identify it?** (this chapter)
3. **Does the estimator converge with the announced distribution?** (Ch. 7–8)
4. **Is the selected decision still guaranteed?** (Ch. 9–10)

A positive answer to one answers none of the next.

What a guarantee is, made formal

Definition 6.1 (Transferable guarantee). A guarantee is transported through a representation when the experiment, the target, the procedure, and their correspondences are specified, and the probabilistic or risk statement is proved to be preserved. Changing the experiment — from independent samples to an adaptive trajectory — requires a result covering the new data regime.

Four computed contradictions

Each example isolates one object that ceases to be justified. All numbers are exact and machine-checked (`verificacion/m_ch6_contraejemplos.py`).

Example 6.2 (Unobserved confounding: same law, different intervention). Let $U \sim \text{Bernoulli}(1/2)$ and the recorded action be $A = U$. Two potential-outcomes models agree exactly on the observed law: in M1, $Y(0) = Y(1) = U$; in M2, $Y(a) = a$. Both give $\mathbb{P}(A = Y = 0) = \mathbb{P}(A = Y = 1) = 1/2$. Yet intervening to set $A = 1$ yields

$$\mathbb{E}[Y(1)] = \begin{cases} 1/2 & \text{in M1} \\ 1 & \text{in M2} \end{cases}.$$

No estimator based on those records distinguishes the two models; adding columns independent of (U, A, Y) changes nothing. DM, IS, and DR do not remove this ambiguity by name; randomization, pertinent measurement, or justified structural restrictions change the problem itself.

Example 6.3 (Failure of mixing: one sample is not a sample). Draw $Z \sim \text{Bernoulli}(1/2)$ once and set $X_t = Z$ forever. The process is stationary, its mean over time never converges to $1/2$, and the events $\{X_0 = 1\}, \{X_k = 1\}$ keep dependence $1/4$ at every lag. If $1/2$ is replaced by an unknown proportion p, the observed law identifies p through $\mathbb{P}(X_0 = 1) = p$, but lengthening one trajectory

repeats the same Bernoulli estimate. Many independent restarts do identify it: the sampling unit changed, not the theorem.

Example 6.4 (Insufficient overlap: identification, not variance). A register shows only action 0 with reward zero; the target policy takes action 1 with probability 0.9. Two models assigning reward 0 and 1 to the unseen action produce identical registers but target values 0 and 0.9: with $0 \leq R \leq 1$, the identification interval is $[0, 0.9]$, and no point estimator is identified by the data. DM extrapolates; that is not an identification earned by the data. Distinct from this is **weak** overlap: $\pi_b(1) = \epsilon > 0, \pi_e(1) = 0.9, R = A$. One IS contribution is $0.9/\epsilon$ with probability ϵ, zero otherwise:

$$\mathbb{E}[W] = 0.9, \quad \mathrm{Var}(W) = 0.81(1/\epsilon - 1),$$

which at $\epsilon = 0.01$ equals 80.19 — finite, not small.

Example 6.5 (Nonstationarity refocuses the target). $n/2$ independent Bernoulli outcomes of mean 0.2, then $n/2$ of mean 0.8. The pooled mean estimates 0.5 — the average of regimes — with variance $0.16/n$, and it admits a CLT for that average. Nonstationarity does not kill inference; it forces a declaration of which population and period are being evaluated.

The L1–L6 failure map

Mode	Object	Operational question
L1	State memory	Do histories collapsed into the same summary require different decisions?
L2	Observed information	What is inferred, under which model, from which history?
L3	Dynamics and sampling	One episode, one trajectory, or a changing regime?
L4	Intervention	Does the observed law determine the outcome of changing the policy?
L5	Support	Is there information in the regions the target policy visits?
L6	Estimator	Do remainder and variance permit the announced distribution?

Table 6.1: The map of six failure mechanisms; each row names the first object to audit.

Example 6.6 (L1: two histories, one summary). At the final decision of an episode, $S_1 = 0$ always, but the reward is $R_1 = 1$ if $A_1 = A_0$ and zero otherwise, with A_0 chosen uniformly. A rule remembering the previous action scores 1; any rule using only S_1 plus independent randomization scores $1/2$. The summary erased decisive information; enlarging the state to (S_1, A_0) restores a Bellman equation for the enlarged problem.

Note 6.7 (Auxiliary-function error rates are joint). For an orthogonal expansion with remainder bounded by Ca_nb_n (errors L^2 of two auxiliaries), the condition is $a_nb_n = o_P\big(n^{-1/2}\big)$, together with the other hypotheses of the expansion — not a universal law. With $a_n = O_P(n^{-0.30})$, $b_n = O_P(n^{-0.25})$ the product is $o_P\big(n^{-1/2}\big)$; two rates of $n^{-0.20}$ each give $O_P(n^{-0.40})$, which does not certify root-n inference. Cross-fitting controls empirical terms; it establishes no learning rates by itself.

When do Bayes, minimax, and admissibility coincide?

In a known model with fixed initial distribution ν, all policies are compared by one scalar $J_\nu(\pi)$: if the maximum is attained, the attaining policies are exactly the admissible ones for loss $-J_\nu$ over a single "parameter", and Bayes and minimax coincide on that scalar. Indexing risk by initial states instead, $R(s,\pi) = -V^\pi(s)$, makes the comparison vectorial; a uniformly optimal stationary policy exists in the finite discounted MDP and dominates everything else, so admissibility and uniform optimality coincide — **but** minimax may admit additional policies:

Example 6.8 (A minimax policy can be dominated). One decision, two initial states, feasible value vectors $V^a = (1, 0)$ and $V^b = (0, 0)$. The first dominates the second, yet $\min_s V^a(s) = \min_s V^b(s) = 0$: both are maximin. Under a prior with positive mass on the first state only a is Bayes; under a prior concentrated on the second, both are. Existence of a uniformly optimal policy does not equate the three solution sets.

With an unknown model family, the prior and the worst case matter again: observing the state does not reveal the model (`verificacion/m_ch6_tricotomia.py`).

The justification file

Note 6.9 (Protocol before invoking a guarantee). Record the target (with horizon, discount, initial population, period), the policy (fixed, learned, or selected during evaluation), the independent unit, the model and which of its parts are designed versus estimated; separate identification assumptions from estimator conditions; cite the applicable result; document which conditions are known by design and which have only diagnostic support. If an essential condition remains unjustified, report a descriptive estimate, an identification region, or a sensitivity analysis — without attaching the guarantee that was not established. Chapter 10 turns this file into algebra.

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Chapter 7

Estimating policy value: DM, IS, DR

Off-policy evaluation (OPE) asks what a rule is worth using records produced by a different rule. This chapter fixes the experiment — independent episodes of a finite horizon — and develops the three estimation routes with their variance analysis. The sampling regime of a single long trajectory belongs to Chapter 8′s references and to the diagnostics of Chapter 6.

The experiment and the assumption set

Definition 7.1 (OPE experiment and target). n i.i.d. episodes

$$O_i = \left(S_{i,0}, A_{i,0}, R_{i,0}, ..., S_{i,H}\right)$$

of fixed horizon H, finite actions; behavior policy $b_t(a|s)$; fixed target policy $e_t(a|s)$ (both time-varying allowed). Common initial law ν; reward/transition kernels $K_t(\mathrm{d}r, \mathrm{d}s'|s,a)$; $|R_t| \le R_*$; $0 < \gamma \le 1$. The target is

$$J(e) = \mathbb{E}_e(G) = \int v_0(s)\nu(\mathrm{d}s), \quad G = \sum_{t=0}^{H-1} \gamma^t R_t,$$

with $q_t(s,a) = \mathbb{E}[R_t + \gamma v_{t+1}(S_{t+1})|S_t = s, A_t = a]$ and $v_t(s) = \sum_a e_t(a|s) q_t(s,a)$. Estimating $v_0(s_0)$ at a fixed state is a different problem.

Assumption 7.1 (Consistency and well-defined intervention). Observed states and rewards equal their potential outcomes under the action sequence actually received; action versions are specified; no interference between episodes. The outcome at t may depend on the whole past sequence.

Assumption 7.2 (Support of the target policy). On histories reachable under e, every action of positive target probability has positive behavior probability. For quantitative bounds we impose the ratio bound $e_t/b_t \le C < \infty$ on the relevant support — strictly stronger than positivity; the largest observed ratio in a sample is a diagnostic, not the population coefficient $\rho_{\max} = \sup e_t/b_t$.

Assumption 7.3 (Exchangeability and Markov structure). Given the pre-action history, the recorded action is independent of future potential outcomes (sequential exchangeability); S_t summarizes that history for kernels and policies. Observational Markovianity does not imply absence of confounding (Example 6.2).

Theorem 7.2 (Identification of policy value). *Under the assumptions above, with* $\rho_t = e_t(A_t|S_t)/b_t(A_t|S_t)$, $W_t = \prod_{k=0}^{t} \rho_k$, $W_{-1} = 1$, *and returns integrable under the target law,*

$$J(e) = \mathbb{E}_b(W_{H-1} G) = \sum_{t=0}^{H-1} \gamma^t \mathbb{E}_b(W_t R_t).$$

Proof. Factorize the episode law into initial distribution, policies, and kernels; in the density ratio, ν and the kernels cancel, leaving W_{H-1}; for the reward at t the history up to that reward suffices, W_t. Section A.5.1. □

Direct method

Definition 7.3 (DM). Fit $\hat{q}_t$ backward through Bellman regressions; with $\hat{v}_t(s) = \sum_a e_t(a|s)\hat{q}_t(s,a)$,

$$\hat{J}_{\mathrm{DM}} = \frac{1}{n}\sum_{i=1}^{n} \hat{v}_0(S_{i,0})$$

over evaluation states independent of the fit (cross-fitting permitted episode-wise).

Proposition 7.4 (DM bias and variability). *Conditionally on an independent fit, the bias is* $\mathbb{E}_\nu(\hat{v}_0 - v_0)$*, bounded in magnitude by* $\mathbb{E}_\nu\left[\sum_a e_0(a|S_0)|\hat{q}_0(S_0,a) - q_0(S_0,a)|\right]$*, and the conditional variance is* $\mathrm{Var}_\nu \frac{\hat{v}_0}{n}$*; total variability includes the fit itself. DM needs no explicit* b *— but it extrapolates into support-free regions only by imposing restrictions the data do not contain.*

Importance sampling

Definition 7.5 (IS and PDIS).

$$\hat{J}_{\mathrm{IS}} = \frac{1}{n}\sum_{i=1}^{n} W_{i,H-1}G_i, \quad \hat{J}_{\mathrm{PDIS}} = \frac{1}{n}\sum_{i=1}^{n}\sum_{t=0}^{H-1} \gamma^t W_{i,t}R_{i,t}.$$

Exact statements below use **true** behavior probabilities, known by design; substituting estimated ones does not automatically preserve finite-sample unbiasedness. The self-normalized variant is a different procedure with finite bias.

Theorem 7.6 (IS unbiasedness). *With true weights and* $\mathbb{E}_e|G| < \infty$*,* $\mathbb{E}_b\left(\hat{J}_{\mathrm{IS}}\right) = J(e)$*; if* $\mathbb{E}_b(W_{H-1}^2 G^2) < \infty$*, then* $\mathrm{Var}_b\left(\hat{J}_{\mathrm{IS}}\right) = \mathrm{Var}_b(W_{H-1}G)/n$*. The second-moment condition is not needed for unbiasedness (Section A.5.1).*

Theorem 7.7 (Variance bound and curse of horizon). *If* $|G| \le B_H$ *and* $\rho_t \le C$*, then*

$$\mathrm{Var}_b\left(\hat{J}_{\mathrm{IS}}\right) \le B_H^2 C^H/n,$$

an upper bound only. The growth it permits is real: in one state, Bernoulli(b) versus Bernoulli(p) actions and terminal reward one,

$$\mathrm{Var}_b\left(\hat{J}_{\mathrm{IS}}\right) = \frac{c^H - 1}{n}, \quad c = \frac{p^2}{b} + \frac{(1-p)^2}{1-b} = 1 + (p-b)^2/(b(1-b)) > 1,$$

by exact multiplication of second moments (`verificacion/m_ch7_curse.py`; Section A.5.2 knows why a bound cannot be improved to a lower one).

Doubly robust estimation

Definition 7.8 (DR, one decision and its sequential form). One decision, $m(s,a) = \mathbb{E}[R|S,A]$:

$$D_i = \hat{v}(S_i) + \frac{e(A_i|S_i)}{\hat{b}(A_i|S_i)}(R_i - \hat{m}(S_i,A_i)), \quad \hat{v}(s) = \sum_a e(a|s)\hat{m}(s,a).$$

The direct term averages actions under e — replacing $\hat{v}$ by $\hat{m}(S_i, A_i)$ destroys the identity even with a perfect model. For H decisions, with models trained outside episode i,

$$D_i = \hat{v}_0(S_{i,0}) + \sum_{t=0}^{H-1} \gamma^t \widehat{W}_{i,t}\big(R_{i,t} + \gamma \hat{v}_{t+1}(S_{i,t+1}) - \hat{q}_t(S_{i,t}, A_{i,t})\big).$$

Theorem 7.9 (Double robustness identity). *For one decision, with fixed candidate functions $\overline{m}, \overline{b} > 0$,*

$$\mathbb{E}_b(D) - J(e) = \mathbb{E}_\nu \left[\sum_a e(a|S) \frac{\overline{b}(a|S) - b(a|S)}{\overline{b}(a|S)} (\overline{m}(S,a) - m(S,a)) \right] :$$

the bias is a product of both errors and vanishes if either auxiliary is right (Section A.5.3; the sequential cancellation is in the same appendix).

Proposition 7.10 (Efficient inference in one decision). *i.i.d. episodes, finite actions, fixed e, uniformly bounded rewards and fits, $b, \hat{b} \geq \epsilon > 0$, episode-wise cross-fitting with a fixed number of folds, both L^2 errors to zero with product $o_P\big(n^{-1/2}\big)$, and positive limiting variance. Then*

$$\sqrt{n}\big(\hat{J}_{\mathrm{DR}} - J\big) = \frac{1}{\sqrt{n}} \sum_i \varphi(O_i) + o_P(1),$$

$$\varphi(O) = v(S) - J + \frac{e(A|S)}{b(A|S)}(R - m(S,A)),$$

the canonical gradient in the nonparametric one-decision model; its variance is the efficiency bound for regular estimators (Section A.5.4). Efficiency is more than double robustness: two $O_P\big(n^{-1/4}\big)$ errors do not by themselves certify the little-o product (Note 6.7).

Note 7.11 (Support states of an evaluation). Identification (assumption set), bounded-ratio regime, and finite-second- moment regime are successive support states of an OPE claim; each weaker state permits proportionally weaker conclusions (regions, then point estimates with heavy-tailed variance, then efficient intervals). Declaring the state is part of the estimand; truncating weights trades bias for variance and must itself be reported — including its threshold and the bias it introduced (Example 6.4 for the endpoint case).

For the time-unconstrained kernel model of finite horizon with unknown initial law, the efficient influence function uses marginal occupancy ratios $w_t = d_t^e / d_t^b$ instead of products W_t — Kallus & Uehara (2020) with its own conditions (Appendix B), independent of the Wald embedding.

Readings

Jiang & Li (2016) and Dudík et al. (2011) for DR in policy evaluation; Precup et al. (2000) for IS via eligibility; Thomas et al. (2015) for high-confidence OPE; Liu et al. (2018) for the curse-of-horizon and breaking it with occupancy ratios; Kallus & Uehara (2020) and Kallus & Uehara (2022) for semiparametric efficiency, each in its own temporal regime.

Chapter 8

Inference while the experiment runs

A fixed-time confidence interval is a promise about one analysis at one sample size. An agent dashboard violates that promise on arrival: the analyst looks every day, stops when the interval excludes zero, and reports that moment's interval. Under optional stopping, the nominal coverage of a fixed-time interval is gone — this is not a subtlety but a different inferential object. **Safe anytime-valid inference** (SAVI) builds the object whose guarantee survives exactly that usage: e-processes for testing and confidence sequences for estimation.

This chapter is self-contained up to Ville's inequality, states the validity theorem for off-policy confidence sequences with full hypotheses and proof, and then points to the general framework of Waudby-Smith et al. (2022) with the conditions its abstract declares. It is the book's main addition over its course-book companion (measured gap: the companion contained none of this material).

E-values and e-processes

Definition 8.1 (E-value). A random variable $E \geq 0$ is an e-value for a hypothesis $\mathcal{P}$ if $\sup_{P \in \mathcal{P}} \mathbb{E}_P[E] \leq 1$. Large observed values are evidence against $\mathcal{P}$: by Markov's inequality, $P(E \geq 1/\alpha) \leq \alpha$ for every $P \in \mathcal{P}$, so E rejects at level α when $E \geq 1/\alpha$. A likelihood ratio $E = q(X)/p(X)$ is an e-value for the singleton $\{p\}$: $\mathbb{E}_p[E] = 1$.

Definition 8.2 (E-process and test (super)martingale). With a filtration $(\mathcal{F}_t)$, an e-process for $\mathcal{P}$ is an adapted sequence $E_t \geq 0$ such that for every $P \in \mathcal{P}$ and every stopping time τ, $\mathbb{E}_P[E_\tau] \leq 1$. A test supermartingale for P is a nonnegative supermartingale starting at 1; every test supermartingale is an e-process for its hypothesis, since optional stopping gives $\mathbb{E}[E_\tau] \leq \mathbb{E}[E_0] = 1$ whenever the optional stopping theorem applies — for nonnegative supermartingales, Fatou's lemma extends it to arbitrary, possibly unbounded, stopping times. Predictable multiplicative factors with conditional expectation at most one are the standard constructor.

Theorem 8.3 (Ville's inequality). *If $(M_t)_{t \geq 0}$ is a nonnegative supermartingale with $M_0 = 1$, then for every $a > 1$,*

$$\mathbb{P}(\exists t : M_t \geq a) \leq \frac{1}{a}.$$

Proof. Let $\tau = \inf\{t : M_t \geq a\}$ (infinite if never). For each finite T, $\tau \wedge T$ is a bounded stopping time, and on $\{\tau \leq T\}$, $M_{\tau \wedge T} \geq a$; hence $a \cdot \mathbb{P}(\tau \leq T) \leq \mathbb{E}[M_{\tau \wedge T}] \leq 1$ by bounded optional stopping. Let $T \to \infty$: $\mathbb{P}(\tau < \infty) = \lim_T \mathbb{P}(\tau \leq T) \leq 1/a$. The event $\{\exists t : M_t \geq a\}$ is $\{\tau < \infty\}$. □

Definition 8.4 (Confidence sequence). A $(1-\alpha)$-CS for θ is a sequence of sets (C_t) with $\mathbb{P}(\forall t \geq 1 : \theta \in C_t) \geq 1 - \alpha$: simultaneous — hence valid at every stopping time, including "look again tomorrow". Inverting the level-α test from an e-process gives a CS: with $E_t(\theta)$ an e-process for each candidate value, $C_t = \{\theta : E_t(\theta) < 1/\alpha\}$.

A constructor with a full proof: the Hoeffding e-process

Definition 8.5 (Conditionally bounded contributions). Let X_t be adapted with $\mathbb{E}[X_t|\mathcal{F}_{t-1}] = \mu$ and $a_t \le X_t \le b_t$ with a_t, b_t predictable and $b_t - a_t \le 1$. For any predictable λ_t define the factor

$$K_t = \exp\Big(\lambda_t(X_t - \mu) - (\lambda_t^2)(b_t - a_t)^2/8\Big)$$

and the process $M_t = \prod_{s \le t} K_s$, $M_0 = 1$.

Theorem 8.6 (Validity of the Hoeffding e-process). *M_t is a nonnegative supermartingale starting at 1, hence an e-process: for every stopping time τ, $\mathbb{E}[M_\tau] \le 1$. Consequently, for i.i.d. X_i bounded in an interval of width w and mean μ,*

$$C_t = \bar{X}_t \pm \sqrt{\frac{w^2\big(\log(2/\alpha) + t\lambda_0^2/8\big)}{t\lambda_0}}$$

optimized over the fixed λ_0 is a valid fixed-λ CS; mixing $K_t(\lambda)$ over a probability distribution F on λ preserves the e-process property by Tonelli's theorem and yields variance-adaptive mixtures — whose tight, nonasymptotic constants are the contribution of Howard et al. (2021) and are quoted there, not reproved here.

Proof. By Hoeffding's lemma, a mean-zero variable bounded in an interval of width d satisfies $\mathbb{E}[e^{\lambda Y}] \le e^{\lambda^2 d^2/8}$; applied conditionally to $X_t - \mu$ with $d = b_t - a_t$, $\mathbb{E}[K_t|\mathcal{F}_{t-1}] \le 1$, so $\mathbb{E}[M_t|\mathcal{F}_{t-1}] = M_{t-1}\mathbb{E}[K_t|\mathcal{F}_{t-1}] \le M_{t-1}$. Nonnegative, starts at 1: the claims follow from Definition 8.2 and Theorem 8.3. The displayed form of the interval comes from solving $\lambda(|x|) - \lambda^2 w^2/8 = \log(2/\alpha)/t$ for $|x|$ after the two-sided union bound. □

Anytime-valid off-policy evaluation

Theorem 8.7 (OPE confidence sequence under a ratio bound). *In the one-decision experiment of Definition 7.8 with i.i.d. episodes, true behavior propensities known by design, target support ratio $w = e/b \le C$ on the target's support, and bounded reward $|R| \le B$, define $X_i = W_i R_i$ with $W_i = e(A_i|S_i)/b(A_i|S_i)$. Then $\mathbb{E}_b[X_i] = J(e)$ (Theorem 7.2), $|X_i| \le CB$, and the process*

$$M_t(\lambda) = \prod_{i \le t} \exp(\lambda(X_i - J(e)) - (2\lambda CB)^2/8)$$

is an e-process for the value hypothesis; the induced $(1-\alpha)$-CS for $J(e)$ is valid at arbitrary stopping times, including continuous monitoring of the logged evaluation.

Proof. Theorem 7.2 gives the mean; boundedness gives the width $2CB$; the factors have conditional expectation at most one by Hoeffding's lemma exactly as in Theorem 8.6, with note: the assumed bound is on the **true** ratio, a designed-support quantity — the largest sample ratio is a diagnostic, not this constant (Assumption 7.2). Estimated weights leave the theorem's hypotheses (Definition 7.5 on true weights); their analysis is the framework below. □

Note 8.8 (The general framework and its declared conditions). Waudby-Smith et al. (2022) derive doubly robust confidence sequences for off-policy mean reward in contextual bandits that remain valid at arbitrary stopping times while: relaxing fixed-sample analysis, uniformly bounded weights, constant logging policies, and constant policy values; allowing the logging policy itself to learn; and contexts forming a dependent time series. The e-processes use predictable weighting so that no weight bound need be known, and estimates adapt to empirical variance. No claim of

that strength is proved in this chapter; Theorem 8.7 is the fully proved backbone in the designed-support regime, and the two are used together in Chapter 11.

Using e-values correctly

1. **Predictability is the contract.** Any quantity entering the factor at t must be fixed before observing X_t; choosing the e-process after seeing the data voids the guarantee, as does reusing the same data to tune its parameters.
2. **Combination rules.** The arithmetic mean of e-values is an e-value ($\mathbb{E}\left[\bar{E}\right] \leq 1$ by linearity — the proof is one line and worth keeping: it licenses running several monitors and combining them without new hypotheses). Products combine e-values exactly under independence, and by the e-variable calculus of Vovk & Wang (2021).
3. **Power and safety trade off.** The e-process for a designed everything freely spendable at $1/\alpha$ buys the right, not the tightness; variance adaptation exists to recover width, cf. Note 8.8.
4. **Stopping to deploy.** A lower CS bound crossing a deployment margin δ_0 at a data-dependent time is a level-α statement even with that continuation rule; the remaining questions — causal scope, population, constraints — are outside any interval (Note 6.9).

Laboratories

`labs/m_ch8_labA_stopping.py`: optional-stopping attack on the fixed-time interval versus the CS, with Monte Carlo error of every reported coverage. `labs/m_ch8_labB_gridcs.py`: mixture-grid CS for OPE value under designed support; widths reported with their seeds. Outputs are registered in `resultados_M.json`; the verification suite `verificacion/m_ch8_anytime.py` rechecks the e-process expectation bound and Ville's threshold numerically.

Readings

Ramdas et al. (2023) for the SAVI program and its game-theoretic language; Vovk & Wang (2021) for e-value calibration and combination; Howard et al. (2021) for nonasymptotic CSs; Waudby-Smith et al. (2022) for off-policy inference with adaptive logging.

DRAFT Statistical Guarantees for Sequential Decisions

Chapter 9

Learned policies: evaluation, gradients, SAVE

Everything so far evaluated a **fixed** policy. When the same data also train the policy, the object evaluated can change with each block — and the error that evaluation must control changes character. This chapter fixes the three pieces that remain rigorous here: the policy-gradient identity with its baseline calculus, the sequential organization of evaluation (SAVE), and the margin condition that converts estimation error into decision loss.

Policy gradient with a computed baseline

Definition 9.1 (Differentiable policy objective). Finite horizon H, fixed initial distribution, environment kernel independent of the policy parameter θ, $J(\theta) = \mathbb{E}_\theta\left(\sum_{t=0}^{H-1} \gamma^t R_t\right)$, action score $z_t = \nabla_\theta \log \pi_\theta(A_t|S_t)$, return-to-go G_t; enough integrability to exchange derivative and integral.

Theorem 9.2 (Policy gradient identity).

$$\nabla_\theta J(\theta) = \mathbb{E}_\theta\left(\sum_{t=0}^{H-1} \gamma^t z_t G_t\right) = \mathbb{E}_\theta\left(\sum_{t=0}^{H-1} \gamma^t z_t q_t^{\pi_\theta}(S_t, A_t)\right).$$

Proof. Only action probabilities depend on θ, so the trajectory score is $\sum_t z_t$; rewards before t vanish in expectation against z_t (conditional mean zero given history); conditioning on (S_t, A_t) substitutes G_t by q_t. Full steps: Section A.6.1. Infinite horizon needs domination conditions: $\gamma < 1$ alone does not control an arbitrarily large score. □

Proposition 9.3 (Baselines: centring and variance). *If b_t depends on the pre-action history but not on the sampled action, $\mathbb{E}(z_t b_t) = 0$ under integrability, and G_t may be replaced by $G_t - b_t$ without changing the gradient's mean. For a vectorial single-time contribution, minimizing the trace of the conditional variance gives*

$$b^*(s) = \frac{\mathbb{E}(\|z\|^2 G | S = s)}{\mathbb{E}(\|z\|^2 | S = s)},$$

when the denominator is positive — which need not equal $V(s)$, and does not solve the many-times minimization automatically.

Example 9.4 *(synthetic)* **(A baseline can help, hurt, or kill variance).** One abstract decision: A Bernoulli with $p = 1/\left(1 + e^{-\theta}\right)$, $R = A$, $p = 0.75$; true gradient $p(1-p) = 0.1875$. Contribution without baseline $A(A - p)$: values in $\{0, 0.25\}$, variance 0.01171875. Baseline $b = p$: values $\{0.5625, 0.0625\}$, variance 0.046875 — four times larger. Baseline $b = 1 - p$: value 0.1875 for both actions, variance zero. Same model, three outcomes (`verificacion/m_ch9_pg.py`).

Example 9.5 *(synthetic)* **(Continuous state, checkable gradient).** S uniform on $[-1,1]$, $\pi_\theta(1|s) = 1/\big(1+e^{-\theta s}\big)$, $R = AS$; at $\theta = 0$ the true gradient is $\mathbb{E}(S^2)/4 = 1/12$. Contribution without baseline has variance $13/720$; with $b(s) = s/2$ it is $1/180$: a reduction of $9/13 \approx 69.2\%$ from $\mathbb{E}(S^2) = 1/3$ and $\mathbb{E}(S^4) = 1/5$ alone (`verificacion/m_ch9_pg.py`).

When the data also train the policy

Let π_k be learned before an evaluation block and $\hat{J}_k$ estimate its value; a weighted average with predetermined weights a_k estimates a mixture of values of **different** policies, and against a final π_f appears the term

$$\sum_k a_k\big(J(\pi_k) - J\big(\pi_f\big)\big), \quad \sum_k a_k = 1,$$

which does not vanish by respecting temporal order even with centred block errors. Controlling it at the interval's scale is the problem SAVE (SequentiAl Value Evaluation) solves by its specific partition, aggregation, and studentization; an arbitrary average of DR contributions does not reproduce it.

Note 9.6 (What may be attributed, and to what). The value-function inference of Shi et al. (2022) obtains, for a fixed policy, a studentized

$$\sqrt{nT}\Big(\hat{J}(\pi;\nu) - J(\pi;\nu)\Big)/\hat{\sigma}(\pi;\nu) \xrightarrow{d} N(0,1)$$

under the complete hypotheses of their Theorem 1 — bases, regularity, information, dimension growth — in their regimes with n or T increasing. Appendix B carries the reading-level statement; their proofs are an outline in the article and full in its supplement. Sequential evaluation under their Teorems 3–4 ties value rates to margin conditions; none of it is a re-statement of DR, none of it follows from the Wald embedding, and learning the logging policy is its own problem.

Example 9.7 *(synthetic)* **(Margin converts error into loss, elementarily).** Two actions, $\Delta(s) = |q(s,1) - q(s,0)|$, and suppose $\mathbb{P}(0 < \Delta(S) \le u) \le Cu^\kappa$. If $\|\hat{q} - q\|_\infty \le \epsilon$, a wrong greedy action has positive gap at most 2ϵ, so

$$\mathbb{E}(q(S,a^*(S)) - q(S,\hat{a}(S))) \le C(2\epsilon)^{1+\kappa}.$$

Ties cost zero. This calculation is self-contained for one decision; it certifies no neural error rate and substitutes none of the article's longitudinal conditions (`verificacion/m_ch9_margin.py`).

Note 9.8 (Freeze, then evaluate). With a policy learned and frozen on one part of the data, evaluation on independent episodes conditions on the training set: the target is the value of **that** policy — not the unknown optimal one — and Chapters 7–8 apply as stated. A pointwise interval for the learned policy is never a simultaneous band over the whole search.

Readings

Shi et al. (2022) with its supplement before any use of its normalization; Sutton & Barto (2018) for policy gradients in context; Kallus & Uehara (2020) for what efficiency means in the same temporal model.

DRAFT v0.1-M — Statistical Guarantees for Sequential Decision Agents

Part III

Transfer and verification

A guarantee is an object with premises. Transfer certificates make those premises algebraic; a real-data study executes the protocol end to end; the frontier asks what survives when actions are text and policies are unobserved.

Chapter 10

Transfer certificates

A guarantee is an object with premises. Chapter 6 produced the vocabulary to locate failure; this chapter turns it into a small algebra. The claim is modest and exact: a **certificate** is a tuple with a rule for when it discharges a guarantee; composition is permitted by three proved rules; and three computed counterexamples mark the compositions that fail. The output is a soundness criterion — **what may be reported** — that the rest of the book then applies to itself, including to the study of Chapter 11.

The object

Definition 10.1 (Transfer certificate). A certificate is a tuple $\mathcal{C} = (K, \mathcal{R}, \mathcal{K}, \sigma)$: a claim K about a statistical object, its data regime $\mathcal{R}$ (experiment and sampling unit), a finite set of conditions $\mathcal{K}$ (identification, estimator, variance — each with its formal statement), and a status map $\sigma : \mathcal{K} \to \{\text{design}, \text{proved}, \text{diagnostic}, \text{open}\}$. A certificate **discharges** K at level $\frac{\text{design}}{\text{proved}}$ when every condition has status design or proved; a condition at diagnostic or open caps the reportable conclusion accordingly (descriptive statement, identification region, or sensitivity analysis). Two certificates are **compatible** when they are stated for the same regime and their condition sets agree on every shared hypothesis.

Note 10.2 (Diagnostics are not conditions). ESS ratios, autocorrelation plots, small Bellman residuals, stable numerics: they populate the diagnostic status and nothing else. No threshold of any of them discharges overlap, mixing, or identification; those are population statements (Note 6.7, Example 6.3).

Three composition rules, proved

Proposition 10.3 (R1 — Image transport of sets). *If C_n covers θ with probability at least $1 - \alpha$ in regime $\mathcal{R}$, then $g(C_n)$ covers $g(\theta)$ at the same level. Coverage may increase; the image need not be an interval — taking its interval hull preserves the inequality when well-defined. Proof: the event $\{\theta \in C_n\}$ is contained in $\{g(\theta) \in g(C_n)\}$. The rule is exact and requires no new condition; it never produces a normal interval by itself.*

Proposition 10.4 (R2 — Chained limit with a linking condition). *A certificate for $\sqrt{n}\left(\hat{\theta} - \theta\right) \xrightarrow{d} N(0, \Sigma)$ and a delta-method certificate compose into one for $g\left(\hat{\theta}\right)$ **if** g is differentiable at θ with nonvanishing gradient on the support of the limit (and Slutsky-consistent variance estimation is discharged in the same regime), giving variance $\nabla g(\theta)^T \Sigma \nabla g(\theta)$. The link condition is the differentiability with nondegenerate gradient; R2 fails at points where it fails (Example 10.6). Proofs of the two ingredients are in Appendix B; composition is the statement made here (`verificacion/m_ch10_certificados.py` recomputes both failure examples).*

Proposition 10.5 (R3 — Hypothesis transport is exact only with the rejection function). *Testing $g(\theta) = v_0$ means testing the composite set $\{\theta : g(\theta) = v_0\}$; it coincides with $\theta = \theta_0$ only if the relevant preimage is that point. Level and power are preserved exactly when the experiment and the rejection function are preserved under a correspondence of hypotheses; changing the data regime (e.g., to dependent trajectories) requires a new justification of the statistic's law.*

Three computed counterexamples to naive composition

Example 10.6 (Efficiency does not paste through zeros of the gradient). X_i i.i.d. $N(\theta, 1)$; the mean is efficient for θ. For $g(\theta) = \theta^2$ with $\theta \neq 0$, $\overline{X}^2$ has asymptotic variance $4\theta^2/n$; at $\theta = 0$, $n\overline{X}^2 \sim \chi^2_1$: the scale and the limiting law change, and a normal formula patched with a zero derivative yields no valid interval. Efficiency is declared for a model, a functional, and a regularity notion — R2′s link condition is not decoration.

Example 10.7 (A preimage with two points). For $g(\theta) = \theta^2$, the hypothesis $g(\theta) = 1$ contains $\theta = 1$ and $\theta = -1$; a test built only for $\theta = 1$ does not control level over the composite hypothesis (R3).

Example 10.8 (Interval for a selected policy). A pointwise $(1 - \alpha)$ interval for a fixed policy, applied to a policy **chosen** because its interval looked best, loses its nominal meaning: selection entered the procedure. The valid object is an evaluation on independent episodes, a uniform band over the search class, or selective inference with its own certificate — three different regimes, three different certificates (Note 9.8).

The soundness rule

Definition 10.9 (Reportability). A guarantee may be reported (with its level) when a discharged certificate for it exists, and it must then be reported with regime $\mathcal{R}$ and the weakest-link status of its conditions. Otherwise the supported report is the degraded one of Definition 10.1. This is a rule about claims, not about effort: a beautiful estimator with an open condition reports less than a plain one fully discharged.

DRAFT Version — Practical Guidelines for Sequential

Claim	Regime	Key conditions	Where proved
$J_\theta(\pi_\delta) = -R(\theta, \delta)$	One fold, finite	Bounded L, same rule class	Theorem 4.3, App. A
Bayes/minimax/admissibility travel	Same	Same class both sides	App. A
Observe-stop recursion optimal	Finite Bayesian	Prior, channel, finite T	Theorem 5.1, App. A
No folding with controlled channel	Sequential, finite	Reachability of the witness	Theorem 4.9, App. A
$\mathbb{E}_b\left(\hat{J}_{\mathrm{IS}}\right) = J(e)$	i.i.d. episodes	Assumptions A1–A3, true weights	Theorem 7.6, App. A
DR bias is a product of errors	One decision	Integrability, fixed candidates	Theorem 7.9, App. A
Efficient expansion of DR	One decision	Cross-fitting, rates, positivity	App. A sketch
CS valid under monitoring	Designed support	Ratio bound C, true weights	Theorem 8.7, App. A
OBD study conclusions	Logged bandit	Per Note 6.9 file	Ch. 11

Table 10.1: Certificates of this book's own guarantees (canonical fragment; status as of this draft — all discharged conditions have their proofs in Appendix A or in the cited sources)

Readings

The calculus is this book's own; its ingredients are standard and credited in line. The certificate format operationalizes the verification discipline of Kallus & Uehara (2020) and the protocol file of Chapter 6; the failure examples are exact computations re-checked by `verificacion/m_ch10_certificados.py`.

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Chapter 11

A real-data study, executed by the protocol

Every preceding chapter made promises. This one executes them, end to end, on data the authors did not generate. The dataset is a logged-bandit subsample of the Open Bandit Dataset (OBD), registered on the fashion e-commerce platform ZOZOTOWN and bundled with version 0.5.7 of the `obp` source distribution (Saito et al. (2021) — the full campaigns download from ZOZO after a use form; the packaged subsamples analyzed here carry 10,000 rounds per behavior policy, campaign `all`). The protocol file precedes the analysis; every number in this chapter is recomputed by `labs/m_ch11_obd.py` and registered in `resultados_M.json`.

Mode L3 of the failure map applies from the start: a packaged subsample is not the full campaign. Conclusions are proportional — this is a real-data study of a protocol, at the scale the packaged data allow.

Protocol file (fixed before the analysis)

Item	Declaration
Estimand	$J(e)$: expected click of the target policy e, one decision per logged round.
Intervention	e chooses the **item**; the logged position mechanism is kept (positions are features of the mechanism, not of the policy). Stated here because it defines the estimand.
Unit	One logged round; the packaged subsample is treated as i.i.d. rounds (the campaign design samples sessions; dependence within session is not accessible in this cut — declared open condition, conclusions capped at "independent rounds").
Target e	ε-greedy, $\varepsilon = 0.5$, over the argmax of a click model fitted on the **separate bts campaign**, frozen before the evaluation log ($K = 80$ items: $e(a^*) = 0.50625$, $e(a) = 0.00625$; ratio bound $C = 40.5$ by construction, not estimated).
Behavior b	The `random` campaign (uniform $\frac{1}{80}$). Recorded propensities: verified in data — **all** 10,000 rows show propensity 0.0125; the bts campaign's recorded propensities span $[4.5 \times 10^{-5}, 0.95424]$. Overlap is exact by design and checked, not assumed.

Item	Declaration
Estimators	DM, IS (true recorded weights), DR (2-fold cross-fitted logistic regressions), and the mixture Hoeffding CS of Theorem 8.7 with $C = 40.5$, $B = 1$.
Variance and CS	Cross-fitted sample variance for DR (asymptotic, Proposition 7.10 regime); the CS is valid at arbitrary stopping times under the same designed support.
Diagnostics	ESS reported as diagnostic only (Note 10.2); no threshold asserted.
Sanity references	On-policy click means of both campaigns (0.0042 bts, 0.0038 random): values of **other** policies, not ground truth for $J(e)$.

Table 11.1: The justification file of this chapter, as Chapter 10 requires

What was verified by design, and what is only diagnosed

Three conditions of Theorem 8.7 are discharged **by design**: i.i.d. rounds (declared), true weights (recorded propensities, exact in the data), and ratio bound $C = 40.5$ (constructed, since e and b are both known deterministic functions of frozen objects). Identification (A1–A3) holds in the campaign design: actions assigned by logged probabilities independent of potential clicks given the recorded context — with the declared caveat that context features here are the packaged ones (hashed user features and affinity scores). The auxiliary-model condition for the DR interval has only model-class support: logistic regressions on affinity and item features have no proved rate in this problem, so the DR interval is read as the asymptotic normal interval of the efficient form, with its own certificate marked partly open. The mean-behavior conditions of the click model behind e themselves are irrelevant to the validity of e as a **target**: target policies need not be good.

Results

Quantity	Value	Reading
n (rounds)	10,000	i.i.d. units as declared
Ratio bound C	40.5	by construction
ESS fraction (diagnostic)	0.048	weights concentrate: only 5% effective rounds drive IS
IS $\hat{J}$	0.0059	true weights; asymptotic CIs inferior at this ESS
DR $\hat{J}$	0.0057	cross-fitted
DM $\hat{J}$	0.0051	model-based
DR normal CI 95%	$[-0.0022, 0.0136]$	includes zero: at binary 0.4% click rates and this ESS, $n = 10^4$ does not separate the target's lift from none
CS 95% at n	$[-0.2, 1.0]$ — vacuous	the nonasymptotic mixture Hoeffding CS with $d = 2C = 81$ carries width $\sim \sqrt{d^2 \log(1/\alpha)/n}$: honest arithmetic for anyone who plans monitoring on it

Table 11.2: OBD packaged subsamples, campaign `all`: evaluation of the ε-greedy target. All figures recomputed by the lab; intervals as declared in the protocol file

What the study shows, proportionally

1. **The protocol ran.** Every estimator is defensible under a discharged certificate; the chapter needed no informal step between data and report. The one weak link — the parametric click model behind the DR regression — is marked as such, exactly where it lives (Table 11.1).
2. **Overlap dominates inference.** At $C = 40.5$, ESS=4.8% concentrates the weight in the rounds where the target agrees with random logging; the DR interval is honest about the price. A "successful" evaluation at this design would have required a less greedy target (ε closer to 1), and would have estimated a less interesting policy. That trade-off is the design question the book promised to make explicit (Theorem 7.7 is its one-dimensional shadow).
3. **Nonasymptotic validity is expensive.** The CS's vacuity at $n = 10^4$ is not a defect of the method but its price list, printed: variance-adaptive e-processes (Note 8.8) buy back width under the same guarantee. Choosing between a tight asymptotic interval and a nonasymptotic monitoring tool is a declared decision, not an oversight.
4. **Comparisons are between policies, not to truth.** The sanity references show the packaged target's estimated value above both logged policies' means — with an interval that eliminates no alternative. The chapter reports exactly that.

OPEN — Scale-up: rerun the identical pipeline on the full campaigns (download gated by ZOZO's use form) with sessions as units; the packaged subsample does not expose session independence. Same code, larger n, session-level bootstrap or CS — no change of protocol needed.

DRAFT v0.1 — Statistics Guild of Reykjavík — Do not circulate

Chapter 12

Frontier: evaluating language agents

In a language agent, an action is a token sequence whose probability depends on the generation context. The evaluation questions of Part II return with force: what is the action, what is its recorded probability, and what does comparing two texts mean? This chapter states what transfers and what does not, with the partial results proven here marked as such and the rest marked open. Nothing in this chapter claims coverage that the hypotheses do not discharge.

The objects

Definition 12.1 (Language agent). A language agent maps observations $o_t \in \mathcal{O}$ (text, possibly multimodal) into actions $a_t \in \mathcal{A}_L = \bigcup_{k=1}^{L} \Sigma^k$ (sequences of up to L tokens), through a policy

$$\pi(a|o) = \prod_{j=1}^{|a|} p_{\mathrm{LLM}}\big(\mathrm{token}_j \mid o, \mathrm{tokens}_{<j}\big),$$

computed by the **effective sampler** — after temperature, filters, and constraints, whose probabilities may differ from the base model's.

Proposition 12.2 (Propensity from log-probabilities). *If the generator returns per-token log-probabilities of the effective sampler, the logged probability of a complete action (including its termination rule) is computable in log scale as their sum. Limitation: it requires access to the log-probs of the model that generated the data — absent in most commercial deployments and in cross-vendor logs.*

What a decrease in granularity preserves

Definition 12.3 (Template discretization). Fix K templates and a dissimilarity d; map each action to its nearest template with a fixed tie-break. The grouped action space is $\tilde{\mathcal{A}} = \{T_1, \ldots, T_K\}$. Applying OPE requires specifying how an action inside a group is selected and executed, and justifying identification for that intervention.

Proposition 12.4 (A discretization bound needs a declared metric). *In one decision, let $g(a)$ be the executable template and suppose the reward mean satisfies $\sup_{s,a}|m(s,a) - m(s,g(a))| \le \epsilon$. If the grouped policy is the image of the original under g, integrating the reward difference over the initial state gives $\big|J(\pi) - J\big(\pi_g\big)\big| \le \epsilon$. With an analogous uniform condition on Q^π, the performance-difference lemma bounds the discounted infinite-horizon gap by $\epsilon/(1-\gamma)$. Semantic similarity alone establishes neither condition (Section A.7.1).*

Identifiability without the policy's probabilities

Proposition 12.5 (Non-identification without action probabilities). *Consider one-decision evaluation of a fixed target policy e whose action probabilities are **not** computable for any logged or*

novel observation (text out, probabilities unreachable), with contexts s and bounded rewards. Then $J(e)$ is not point-identified from records (S, A, R) alone: $J(e)$ is an integral against the unknown law $e(a|s)$, and two laws for e that agree on all logged support but differ on never-logged texts induce different values of $J(e)$ with identical records. Point estimation is restored only by quantities the record must carry — log-probs (Proposition 12.2), a registered sampling wrapper, or a class constraint on e — and partial identification under explicit rarity models is the remaining honest object (Section A.7.2 discusses the construction).

Note 12.6 (DR conditions for textual actions). For one-decision evaluation with finitely many templates, Proposition 7.10 applies if the grouping defines an identified intervention, the target is fixed, units are i.i.d., rewards and fits are uniformly bounded, propensities (true and fitted) are bounded away from zero on the target support, folds split by independent unit, both auxiliaries converge in L^2 with product $o_P\left(n^{-1/2}\right)$, and the limiting variance is positive. Sequential textual trajectories need the longitudinal estimator and its own theorem; the three single-step requirements in isolation do not prove that extension (certificate status of each condition: to be discharged per deployment, per Chapter 10).

Note 12.7 (Report when the guarantee is absent). When the policy's probabilities are not computable, when the grouping changes the estimand without an approximation bound, or when prompt-history confounding is absent from the state, the correct report is the absence of the guarantee — an identification region, a sensitivity analysis, or a descriptive comparison — not a confidence interval with assumed coverage (Definition 10.9).

OPEN — Open development (planned contribution, not claimed here): a partial-identification calculus for textual actions with rarity models, and a wrapper protocol that makes commercial samplers record effective propensities by construction. Will graduate from this box when the proofs exist.

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Chapter 13

Open problems

Deployment leaves a question open whenever a condition that held the analysis changes. This closing chapter collects the working problems in the form the book has used throughout: what is missing, what would close it, and what a solution has to respect.

Identification under unobserved confounding

The same observed law can admit different interventional values (Example 6.2). Works: partial identification (regions instead of points), sensitivity parameters with honest ranges, auxiliary mediators that restore identifiability (Shi et al. (2024) for confidence intervals in confounded MDPs under mediator assumptions — conditions to read in the original before any use). A solution must respect: a sensitivity parameter is never identified by the same data it protects against.

Observability and filtering

Theorem 5.1 runs on a known channel with a controlled-free observation law. With latent evolving states, the filter is part of the model, and its error enters the certificates. Works: belief-MDP planning with explicit filters; robust policies to filter misspecification; for statistical guarantees, certificates whose conditions include filter error as a discharged or open component. A solution must respect the frontier of Theorem 4.9: control enters the observation kernel and folding dies.

Evaluation without the policy

The frontier of Chapter 12: text actions, unrecorded probabilities, nonstationary generators. Works: wrappers recording effective propensities; rarity models yielding bounds instead of points; grouping with declared metrics. A solution must respect: no guarantee is reported where no certificate discharges it (Definition 10.9); semantic similarity creates no identification (Proposition 12.4).

Dependence and scale

Mixing conditions are population statements that diagnostics cannot discharge (Example 6.3); continuous high-dimension weakens every spectral argument. Works: martingale structures with their own CLTs; e-processes under dependence (predictable constructions are the current frontier); designs that replace assumptions by randomized assignments known by construction. A solution

must respect: choice of unit decides what inference is possible — the episode, the session, or the regime (Table 6.1, mode L3).

Verification of the book's own claims

Phase 0 of the Lean verification plan has already machine-checked the three core identities of Chapter 4 (`VERIFICACION_LEAN_PLAN.md`). Works: extend machine checking to the sequential stopping recursion, the IS and DR identities in finite spaces, and the certificates' composition rules. A solution must respect: formalization fixes the hypotheses as much as the conclusions — the certificates of Chapter 10 are the natural inventory to feed it.

What to read next

Lattimore & Szepesvári (2020) for bandit theory beyond the pointers here; van der Vaart (1998) for asymptotic statistics; Waudby-Smith et al. (2022) for the operating frontier of adaptive evaluation; and the annotated literature file of the companion course book (`LITERATURA_2022_2026.md`) for working annotations per article.

AFI vol. 1 — Statistical Guarantees for Sequential Decision Agents

DRAFT v0.1-M — Statistical Guarantees for Sequential Decision Systems

Appendices

Proofs support every result used in the body; the tools appendix is reference only, organized by what the rest of the book invokes. The synthetic SSCH thread is consolidated for exercises and replication.

Appendix A

Proofs

Proofs referenced from the body, by topic. Statements marked "sketch" declare it on line; every other proof is complete. The numbering in the body points here through its theorem numbers.

Sufficiency and exponential families

A.1.1 Neyman–Fisher factorization

Proof of Theorem 1.3. In the dominated model, choose a countable mixture $Q = \sum_j c_j P_{\theta_j}$, $c_j > 0$, dominating every law of the family (Halmos–Savage lemma, valid in a dominated experiment), and write $f_\theta = \mathrm{d}P_\theta / \mathrm{d}Q$. If T is sufficient, its common conditional kernel $K(t, B)$ is also the law of $X|T$ under Q; with $g_\theta = \mathrm{d}P_\theta^T / \mathrm{d}Q^T$, disintegration gives $P_\theta(B) = \int_B g_\theta(T(x))\,\mathrm{d}Q(x)$, whence $f_\theta(x) = g_\theta(T(x))$ Q-a.s., which is the factorization after multiplying by the density of Q. Conversely, that factorization makes $\mathrm{d}P_\theta / \mathrm{d}Q$ a function of T, so

$$\mathbb{E}_\theta[\mathbb{1}_B|T] = \mathbb{E}_Q[f_\theta \mathbb{1}_B|T] / \mathbb{E}_Q[f_\theta|T],$$

a function of T independent of θ wherever the denominator is positive; outside that support the version is irrelevant under P_θ. Standard Borel spaces provide the regular kernels used. □

A.1.2 Basu's theorem

Proof of Theorem 1.6. Fix a measurable set B for the ancillary statistic and set $h(T) = P_\theta(A \in B|T)$. By sufficiency h has a version free of θ; by the tower property $\mathbb{E}_\theta[h(T)] = P(A \in B)$ for every θ; so $g(T) = h(T) - P(A \in B)$ has mean zero at every parameter and completeness kills it: $h(T) = P(A \in B)$ a.s. — independence. Conditions: T boundedly complete sufficient, A ancillary, conditional probabilities well-defined. □

A.1.3 James–Stein risk identity

Proof of the identity in Example 1.7. Write $\delta_{\mathrm{JS}}(x) = x + g(x)$ with $g(x) = -(p-2)x/\|x\|^2$, $x \neq 0$. Stein's identity for squared error gives $R(\theta, \delta_{\mathrm{JS}}) = p + \mathbb{E}_\theta[\|g(X)\|^2 + 2 \div g(X)]$, and for $x \neq 0$, $\|g\|^2 = (p-2)^2/\|x\|^2$ and $\div\,(x/\|x\|^2) = (p-2)/\|x\|^2$, so $R(\theta, \delta_{\mathrm{JS}}) = p - (p-2)^2 \mathbb{E}_\theta[1/\|X\|^2]$. The expectation is finite for $p \geq 3$ (integrability of $\|x\|^{-2}$ near the origin requires $p > 2$), and the risk of $\delta_0(X) = X$ is p. □

Wald criteria

A.2.1 Posterior minimization, in detail

Proof of Theorem 2.5. Each posterior loss $\ell_\nu(x, a)$ is measurable and finite. Enumerate the actions and pick the first index attaining the minimum; its selection region is a finite intersection of

comparisons of measurable functions, so the rule is measurable. Iterated expectation writes the Bayes risk of any randomized rule as the expectation of a convex combination of posterior losses, each at least $\min_a \ell_\nu(X, a)$; the constructed rule attains the bound. Equality for another rule forces the nonnegative difference to vanish predictively almost surely, hence its mass on minimizers. Compact or noncompact action sets need extra existence and measurable-selection hypotheses not asserted here. □

A.2.2 Minimax certified by Bayes

Proof of Theorem 2.6. For any rule δ, $\sup_\theta R(\theta, \delta \geq \rho_\nu(\delta) \geq \rho_\nu(\delta_\nu)$. By the theorem's condition, $\rho_\nu(\delta_\nu) = \sup_\theta R(\theta, \delta_\nu)$; chaining yields $\sup_\theta R(\theta, \delta) \geq \sup_\theta R(\theta, \delta_\nu)$ for every δ: δ_ν is minimax. The middle inequality needs control of the risk over the **whole** parameter space; constancy on the prior's support alone gives the wrong last step. □

A.2.3 Essentially complete class, finite problem

Proof of Theorem 2.7. With parameters, data, and actions finite, randomized rules form a finite product of compact simplices, and the risk vector map is linear and continuous: its image $C \subset \mathbb{R}^d$ is compact and convex. For an admissible $r \in C$, the open convex set $U = \{u : u_j < r_j \text{ for all } j\}$ is disjoint from C; separation gives $w \neq 0$ with $w^T c \geq w^T u$ for all $c \in C, u \in U$, necessarily $w_j \geq 0$ (unboundedness downward), and $w^T c \geq w^T r$; normalizing produces a prior under which r minimizes Bayes risk. For any r_0, minimize $\sum_j r_j$ over the nonempty compact $C \cap \{r \leq r_0\}$: the minimizer cannot be dominated and is Bayes by the previous argument — essential completeness, with zeros allowed in the prior. A better prior average does not imply coordinate dominance; the proof uses separation and compactness, not that false implication. □

Dynamic programming

A.3.1 Bellman expectation identity

Proof of Theorem 3.4. With $G_t = R_t + \gamma G_{t+1}$ and the Markov property, $V^\pi(s) = \mathbb{E}^\pi[R_t + \gamma G_{t+1} | S_t = s]$ splits by the tower property into the expected immediate reward plus $\gamma \mathbb{E}^\pi[V^\pi(S_{t+1}) | S_t = s]$; conditioning on the action and using the kernel yields the displayed equation. Conditions: finite or countable state space, finite actions, $\gamma < 1$, bounded rewards (the geometric tail bounds the exchange of sum and expectation), valid kernels. □

A.3.2 Contraction and fixed points

Proof of Theorem 3.6. For a fixed s, let a^* attain $(T^*V)(s)$; then $(T^*V)(s) - (T^*W)(s) \leq \gamma \sum_{s'} P(s'|s, a^*)(V(s') - W(s')) \leq \gamma \|V - W\|_\infty$; repeating with the roles exchanged bounds the absolute difference. Averaging over actions preserves the bound for T^π. Conditions as stated; completeness of bounded functions in sup norm is the completeness of uniform convergence — Banach's theorem applies and gives the two unique fixed points of Corollary 3.7. □

A.3.3 Existence and uniqueness of the optimal value

Proof of Theorem 3.8. The fixed point V^* exists and is unique (previous proof). For any stationary π, $T^\pi V^* \leq T^* V^* = V^*$; monotonicity and iteration give $V^\pi \leq V^*$. For policies that may depend on history, the Bellman inequality holds conditionally before each action; iterating it through h decisions yields $\mathbb{E}_\pi\left[\sum_{t<h} \gamma^t R_t + \gamma^h V^*(S_h) | S_0 = s\right] \leq V^*(s)$, and the last term vanishes as h grows (V^* bounded, $\gamma < 1$). A greedy stationary policy at V^* attains equality because its operator has the same fixed point; ties may multiply optima, not the value. □

A.3.4 Policy improvement

Proof of Theorem 3.11. Let $d = T^{\pi'}V^{\pi} - V^{\pi} \geq 0$ for π' greedy at V^{π}; then $V^{\pi'} - V^{\pi} = \left(I - \gamma P^{\pi'}\right)^{-1} d = \sum_{j \geq 0} \gamma^j \left(P^{\pi'}\right)^j d \geq 0$; with the tie-keeping rule a changed action makes $d(s) > 0$ somewhere, and iteration visits no policy twice among finitely many; at a fixed point $T^*V^{\pi} = V^{\pi}$, so the policy is optimal by Theorem 3.8. □

A.3.5 Residual certificates

Proof of Corollary 3.10. Write $\delta = \|V - V^*\|_\infty$, $d = \|V^{\pi_V} - V^*\|_\infty$, with π_V greedy at V so $T^{\pi_V}V = T^*V$. Fixed points and contraction give $d \leq \gamma\delta + \gamma\|V - V^{\pi_V}\|_\infty \leq \gamma\delta + \gamma(\delta + d)$, hence $(1 - \gamma)d \leq 2\gamma\delta$; and with $e = \|T^*V - V\|_\infty$, $\delta \leq e + \gamma\delta$, so $\delta \leq e/(1 - \gamma)$ and $d \leq 2\gamma e/(1 - \gamma)^2$. Requires $\gamma < 1$; the policy bound is zero when $\gamma = 0$. □

The representation and its frontier

A.4.1 Representation, in detail

Proof of Theorem 4.3. As in the body: the maps are mutual inverses on the decision states, and $J_\theta(\pi_\delta) = \sum_x P_\theta(x) r_\theta(x, \delta(x)) = -\sum_x P_\theta(x) L(\theta, \delta(x)) = -R(\theta, \delta)$, with absolute convergence from boundedness and no future rewards by the terminal transition. No statement is made — or needed — about sequential problems in general. □

A.4.2 Criteria transfer

Proof of Corollary 4.5, Corollary 4.6, and Theorem 4.7. Integrating the identity with a prior gives $J_\nu(\pi_\delta) = -\rho_\nu(\delta)$; negating reverses extrema and their sets. For minimax, $\sup_\pi \inf_\theta J_\theta(\pi) = \sup_\pi(-\sup_\theta R(\theta, \delta_\pi)) = -\inf_\delta \sup_\theta R(\theta, \delta)$ by the bijection. Dominance compares weak/strict inequalities over θ; negating turns risk-dominance exactly into return-dominance. Classes are enlarged consistently on both sides when randomization is allowed. □

A.4.3 The structural frontier, in detail

Proof of Theorem 4.9 — boundary discussion. The body's proof is exact as stated. Boundary remarks. (i) The hypothesis needs the witness history reachable: differing transition laws only on null histories can be tolerated by versions (the experiment can collapse null events), which is why reachability is assumed. (ii) The negative claim is about a **single** family $\left(\tilde{P}_\theta\right)$ with strategy-class preservation: folding is always possible after enlarging the index set to include the strategy itself — the trivial, uninteresting representation that returns the original MDP family. (iii) The two admissible folds of Proposition 4.10 are exactly where the obstruction vanishes: no future observation exists (horizon one), or its law does not depend on the action (uncontrolled stopping, handled at information states). □

A.4.4 Information states versus sufficiency

Proof of Note 5.3. For X_1, X_2 i.i.d. Bernoulli(θ), $0 < \theta < 1$, the sequence is Markov, yet $P_\theta(X_1 = 1 | X_2) = \theta$ depends on θ: X_2 is not sufficient. The control-relevant condition for a summary $Z_t = f_t(H_t)$ is that the law of (R_t, Z_{t+1}) given (H_t, A_t) depend only on (Z_t, A_t) for every admissible action, plus a measurable update and a compatible criterion; this is what (k, b) satisfies in Theorem 5.1, by the predictive-law and update computations in its proof clause (i). □

Off-policy estimation

A.5.1 IS unbiasedness and PDIS

Proof of Theorem 7.2 and Theorem 7.6. Factorize the episode measure into initial law, policies, and environment kernels; in the density ratio the environment factors cancel, leaving W_{H-1}, and analogously W_t on the history ending after reward t. Then $\mathbb{E}_b[W_t R_t] = \mathbb{E}_\pi[R_t]$ and $\mathbb{E}_b[W_{H-1} G] = \mathbb{E}_\pi[G] = J(\pi)$ under integrability; the sample mean has the same expectation; second moments, when finite, give the variance formula. True propensities are used; substitutions are another theorem. A causal reading further needs identification of the intervention mechanism (A1–A3 needs of Definition 7.1). □

A.5.2 The curse example

Proof of Theorem 7.7. $\mathbb{E}_b(W) = 1,\ 0 \le W \le C^H$, so $\mathrm{Var}\big(\hat{J}_{\mathrm{IS}}\big) \le B_H^2 C^H / n$ − upper bound only: with all rewards zero the variance is zero, so no positive lower bound exists uniformly. For the growth example: one-step ratio second moment $c = p^2/b + (1-p)^2/(1-b) = 1 + (p-b)^2/(b(1-b)) > 1$; i.i.d. actions give $\mathbb{E}_b(W^2) = c^H$ exactly, and $\mathrm{Var} = (c^H - 1)/n$ with constant terminal return. □

A.5.3 Double robustness, one decision and sequential

Proof of Theorem 7.9. One decision, candidates fixed with positive denominators on the target support: condition on S and average the action under b to get

$$\mathbb{E}[D|S] = \sum_a \Big[e(a|S)\overline{m}(S,a) + \big(e(a|S)/\overline{b}(a|S)\big) b(a|S)(m(S,a) - \overline{m}(S,a)) \Big];$$

subtracting $\sum_a e(a|S) m(S,a)$ groups the errors into $\sum_a e(a|S)\big(\big(\overline{b} - b\big)/\overline{b}\big)(\overline{m} - m)$, the displayed product form; it vanishes if $\overline{b} = b$ or $\overline{m} = m$. Sequentially, with all q_t correct each residual has conditional mean zero for any integrable candidate weight; with all weights correct, the change of measure telescopes the sum of values to J (with $\overline{v}_H = 0$), the same cancellation written for the finite-horizon estimator of Definition 7.8. Learned models need external training and their own rates; one correct single time does not suffice for the episode. □

A.5.4 Efficient expansion, one decision

Sketch under the theorem's sufficient conditions. With $D_0 = J + \varphi_{\mathrm{eff}}$ and fold means P_k, write

$$\hat{J} - J = (P_n - P) D_0 + \sum_k (n_k/n)\big((P_k - P)\big(\hat{D}_k - D_0\big) + P\big(\hat{D}_k - D_0\big)\big).$$

Conditional on a fold's training, the centered empirical term has variance at most $\big\|\hat{D}_k - D_0\big\|_2^2 / n_k$; boundedness, positivity, and consistency make that norm $o_P(1)$, and a fixed number of folds with Chebyshev gives an $o_P\big(n^{-1/2}\big)$ term without cross-fold independence. The bias identity of Section A.5.3 and Cauchy–Schwarz control the last term by the product of errors, also $o_P\big(n^{-1/2}\big)$ by hypothesis. The i.i.d. CLT on φ_{eff} and the consistent empirical variance complete the expansion. This is the argument's scheme with its conditions named; the full technical verification is textbook semiparametrics. □

Learned policies

A.6.1 Policy gradient identity

Proof of Theorem 9.2. The trajectory log-density under θ has derivative $\sum_t z_t$ (only action probabilities depend on θ, by Definition 9.1's kernel independence). Multiplying by the total return and exchanging derivative and expectation gives $\nabla_\theta J = \mathbb{E}_\theta\big[\big(\sum_t z_t\big) G_0\big]$; each score multiplies only

rewards from its time onward, because $\mathbb{E}_\theta\left[z_t\left(\sum_{s<t}\gamma^s R_s\right)|\mathcal{H}_t\right]$ has conditional mean $\mathbb{E}_\theta[z_t|\mathcal{H}_t] = 0$ times a constant — zero. Weighting by γ^t and conditioning on (S_t, A_t) replaces G_t by q_t, via the tower property inside the expectation. A learned critic $\hat{q}_t$ injects error $\mathbb{E}_\theta\left[\sum_t \gamma^t z_{t(\hat{q}_t - q_t)}\right]$, nonzero in general (Definition 9.1 in Chapter 9). □

Textual actions

A.7.1 Discretization bound

Sketch of Proposition 12.4. One decision: $\left|J(\pi) - J\left(\pi_g\right)\right| = \left|\int \sum_a \left[\pi(a|s)m(s,a) - \pi_g(a|s)m(s,g(a))\right]\nu(\mathrm{d}s)\right| \leq \epsilon$ by the uniform condition and the image-policy construction. With $\sup_{s,a}|Q^\pi(s,a) - Q^\pi(s,g(a))| \leq \epsilon$, the performance-difference lemma telescopes the discounted gap into per-time conditions, each at most $\epsilon\gamma^t$, summing to $\epsilon/(1-\gamma)$. Neither hypothesis follows from embedding similarity of texts. □

A.7.2 Non-identification construction

Proof of Proposition 12.5. With records (S, A, R) only and e unconstrained on never-logged texts, let e_1, e_2 assign different probabilities to a text a^* never in support of the behavior record but of positive record-context probability. The observable law is untouched; the values differ by $(e_1(a^*|s) - e_2(a^*|s))m(s,a^*) \neq 0$ whenever $m(s,a^*) \neq 0$, which is compatible with every logged record. Hence no point identification without carrying the action probability or a class restriction. Bounds from rarity models are the identified object that remains. □

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Appendix B

Tools, referenced by use

Reference only: each entry names the result, its conditions, and where the body invokes it. Statements not proved in Appendix A are attributed.

Exponential families

Note B.1 (Log-partition properties). For $f(x|\eta) = h(x)\exp\{\eta^\top T(x) - A(\eta)\}$ with open natural space: A is convex (Hölder); $\nabla A = \mathbb{E}_\eta[T(X)]$ and $\nabla^2 A = \mathrm{Cov}_\eta(T(X))$ by differentiating under the integral (regularity); the Hessian is positive definite in minimal families; completeness of T holds when the parameter space contains an open set (proof in the course book's appendix; invoked here in Chapter 1).

Asymptotics

Note B.2 (Lindeberg–Lévy CLT). i.i.d. with mean μ and $0 < \sigma^2 < \infty$: $(S_n - n\mu)/(\sigma\sqrt{n}) \xrightarrow{d} N(0,1)$ (characteristic functions; conditions are only those). Delta method: g differentiable with nonvanishing gradient at μ transports the limit with variance $\nabla g^T \Sigma \nabla g$; the vanishing-gradient point is exactly where Example 10.6 stops the naive use. Bootstrap for the mean: with $0 < \sigma^2 < \infty$, the triangular-array conditional CLT under the Lindeberg condition justifies percentile bootstrap CIs of the mean; studentization requires $\sigma^2 > 0$ (conditions in the course book, where they are proved).

Note B.3 (Mixing CLT and one spectral bound). Strictly stationary, mean zero, $\mathbb{E}|X_0|^{2+\delta} < \infty$ for some $\delta > 0$, and $\sum_k \alpha(k)^{\delta/(2+\delta)} < \infty$ (strong mixing on past/future σ-algebras) imply $\sqrt{T}\big(\bar{X}_T\big) \xrightarrow{d} N(0, \sigma^2_{LR})$ with $\sigma^2_{LR} = \mathrm{Var}(X_0) + 2\sum_{k\geq 1}\mathrm{Cov}(X_0, X_k)$, absolutely summable — the form due to the dependent-process literature, quoted from the standard lecture treatment. A finite irreducible aperiodic chain is geometrically mixing; in general state spaces drift/minorization conditions replace that step. Reversible finite chains satisfy $\big\|P^k(s,\cdot) - \mu\big\|_{\mathrm{TV}} \leq 1/2\sqrt{1/\mu(s) - 1}\, r^k$ with $r = \max_{j\geq 2}|\lambda_j|$ — the modulus matters ($\left(\begin{smallmatrix} 0.01 & 0.99 \\ 0.99 & 0.01 \end{smallmatrix}\right)$ has $\lambda_2 = -0.98$).

Note B.4 (HAC variance estimation). Bartlett-weighted $\hat{\sigma}^2_{LR} = \hat{c}_0 + 2\sum_{k=1}^{K}(1 - k/(K+1))\hat{c}_k$; sufficient consistency route: $\sum_k |c_k| < \infty$, $K \to \infty$, $K/T \to 0$, and $\sum_{k\leq K}|\hat{c}_k - c_k| \to 0$ (the last condition must be proved, not assumed from the CLT). Reference: Newey & West (1987) mixing distinctions: Bradley (2005) bootstrap foundations: Hall (1992).

Testing and multiplicity

Note B.5 (Neyman–Pearson, inversion, BH). With densities p_0, p_1, atoms handled by randomization, the likelihood-ratio test is most powerful at its level, including points where $p_0 = 0$; test-inversion yields confidence sets with the dual coverage. Benjamini–Hochberg controls FDR

at $m_0\alpha/m$ with equality under uniform nulls, under independence of null p-values from the rest (proofs in the course book); the PRDS extension is Benjamini & Yekutieli (2001), not the 1995 argument.

Bandits

Note B.6 (UCB1 and Lai–Robbins). Theorem 3.12 quotes the gap-dependent UCB1 bound of Auer et al. (2002) under rewards in $[0, 1]$. For Bernoulli bandits with means in $(0, 1)$ and a unique best arm, any uniformly efficient policy satisfies

$$\liminf_{T\to\infty} \mathcal{R}_T / \log T \geq \sum_{a:\Delta_a>0} \Delta_a / \operatorname{kl}(\mu_a, \mu^*),$$

Lai & Robbins (1985) the general family replaces kl by an infimum over alternative distributions (Lattimore & Szepesvári (2020), where the change-of-measure proofs live). UCB1 does not attain the constant in general.

Semiparametric OPE

Note B.7 (Kallus–Uehara and SZLS, reading-level). In the finite-horizon i.i.d.-episode model with free per-time kernels and unknown initial law, the efficient influence function of $J(e)$ is the augmentation with marginal occupancy ratios $w_t = d_t^e/d_t^b$; double robustness and its conditions are in Kallus & Uehara (2020). The temporally homogeneous infinite-horizon model has another experiment and another bound: Kallus & Uehara (2022) — conditions not interchangeable. For sieve-based value inference at $\gamma < 1$ and the studentized limit for a fixed policy: Theorem 1 of Shi et al. (2022) with its supplement; the sketch in the course book deliberately delegates all its hypotheses.

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Appendix C

The synthetic SSCH thread

SSCH — a hypertension support scenario — is a controlled laboratory for the book's computations. **Synthetic figures**: none of its numbers are clinical estimates; the point of the thread is that every claim about it can be recomputed exactly (`book_stat_inference_v0/lab00_ssch.py` for the longitudinal variant and `monografia_M/verificacion/` for the displays used in the chapters).

The one-signal decision

$\theta \in \{0, 1\}$ indexes need for intensified follow-up; signal $X \in \{0, 1\}$ with $P_0(X = 1) = 0.2$, $P_1(X = 1) = 0.8$; losses 1 and 4 for the two errors. Used in Example 2.10 (risk table, Bayes and minimax rules, and the randomized minimax at $3/8$ with risk $1/2$) and Example 5.4 (one observation at cost 0.05: total risk $0.43 < 0.7$).

Two-state dynamics

Deterministic transitions among $\{g, b\}$, $\gamma = 0.9$; $V^* = (5, 4)$ with the myopic-optimal comparison in Example 3.14. The lesson repeated across the book: immediate-reward maximization can pay no future cost, and proves nothing about learned models.

Two-visit evaluation and paired comparison

The two-visit case of the course book (Chapter 10 there) produces the contribution table reused for intervals: true values 1.425 (evaluated policy, 0.75) and 0.950 (reference, 0.5); in the displayed realization of $n = 100$ independent episodes, DR normal interval $[1.376, 1.474]$ and paired-difference interval $[0.455, 0.495]$ — paired contributions use within-episode differences, with their covariance included. The example's own caution applies: reproducible from the table, not a coverage claim at $n = 100$, and not clinical evidence.

What the thread is for (and not for)

It lets a reader verify an entire argument — representation, Bellman recursion, DR identity, baseline variance — with a pocket calculator or the shipped scripts. It does not validate any deployment claim; every clinical reading is blocked where the example is stated (`es-sintetico` markers).

Appendix D

Reproduction protocols

Everything in this monograph that can be checked mechanically has a shipped script. The rule: every displayed number has a producer; every producer has a fixed seed; every run is registered.

Layout

- `labs/` — the three chapter laboratories:
  - `m_ch8_labA_stopping.py` — peeking attack on fixed-time CIs vs. the CS (stream of Theorem 8.7′s regime; M=4,000 runs, burn-in 10).
  - `m_ch8_labB_gridcs.py` — mixture-grid CS: whole-path validity, $\mathbb{E}[M_\tau]$, final width (M=3,000 runs).
  - `m_ch11_obd.py` — the OBD study end to end (protocol of Table 11.1; data in `labs/data_obd/`, from the obp 0.5.7 sdist `#cite(<saito2021openbandit>, form: "prose")`, bundling 10,000-round subsamples of campaigns `all`, `men`, `women`).
- `verificacion/` — one checker per chapter for the displays of that chapter (`m_ch2_ssch.py`, `m_ch3_bellman.py`, `m_ch3_thompson.py`, `m_ch5_stopping.py`, `m_ch6_contraejemplos.py`, `m_ch6_tricotomia.py`, `m_ch7_curse.py`, `m_ch8_anytime.py`, `m_ch9_pg.py`, `m_ch9_margin.py`, `m_ch10_certificados.py`).
- `pilot_core_proven.lean` — the machine-checked pilot: the three core identities of Chapter 4, proved by Lean and re-verifiable with `lean pilot_core_proven.lean` (see `VERIFICACION_LEAN_PLAN.md`).
- `resultados_M.json` — registered outputs, seeds, and timings per run.

Environments

Manuscript: Typst 0.15.1 (`typst compile main.typ`). Labs: Python 3.12, the pinned environment of the companion book (numpy 2.5.3, sklearn 1.9.1, pandas 3.1.0 for the OBD lab). The OBD CSVs travel inside the obp sdist; provenance recorded in `monografia_M/fichas/saito2021openbandit.md`.

What a rerun may legitimately change

Seeds fix all random draws of the labs and checkers. Legitimate changes on rerun: timings and hardware strings. No number in the text may change without the text changing: that is the contract of Appendix D, and it is enforced by `verificacion/` re-asserting the displayed figures from first principles rather than trusting `resultados_M.json`.

Appendix E

Notation and bilingual glossary

Core notation

Symbol	Meaning
$\Theta, \mathcal{X}, \mathcal{A}, \mathcal{S}$	Parameters, observations, actions, states
P_θ, P_π, P_b	Observation law; laws induced by target and behavior policies
$L(\theta, a), R(\theta, \delta)$	Loss; risk of a rule
$r_\theta(x, a), J_\theta(\pi), J_\nu(\pi)$	Reward; return under one model; prior-averaged return
V^π, Q^π, v_t, q_t	Value functions
$W_t = \prod_{k \le t} \rho_k, w_t$	Cumulative density ratios; marginal occupancy ratios
$m_b(y), B(b, y)$	Predictive law and Bayes update at belief b
$\ell(b, a), W_k(b)$	Posterior expected loss; remaining minimal risk with k observations left
E_t	E-process (nonnegative, mean at most 1 at any stopping time)
(C_t)	Confidence sequence (uniform-in-time coverage)
σ^2_{LR}	Long-run variance under dependence

Table E.1: Core symbols used across the monograph

English–Spanish working glossary (companion book terms)

English (this monograph)	Español (libro del curso)
transfer certificate	certificado de transferencia / ficha de justificación
information state	estado de información
confidence sequence	secuencia de confianza
labeled synthetic example	ejemplo sintético (`es-sintetico`)
opening (OPEN box)	hueco declarado (`PENDIENTE`)
one-decision MDP	MDP de una decisión

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English (this monograph)	**Español (libro del curso)**
folding	incrustación de una decisión
support states of a guarantee	estados de la garantía

Table E.2: Term correspondence with the Spanish course book

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Bibliography

Auer, P., Cesa-Bianchi, N., & Fischer, P. (2002). Finite-time analysis of the multiarmed bandit problem. *Machine Learning*, *47*(2–3), 235–256. https://doi.org/10.1023/A:1013689704352

Bellman, R. (1957). *Dynamic Programming*. Princeton University Press.

Benjamini, Y., & Yekutieli, D. (2001). The Control of the False Discovery Rate in Multiple Testing under Dependency. *The Annals of Statistics*, *29*(4), 1165–1188. https://doi.org/10.1214/aos/1013699998

Berger, J. O. (1985). *Statistical Decision Theory and Bayesian Analysis* (2nd ed.). Springer. https://doi.org/10.1007/978-1-4757-4286-2

Blackwell, D. (1951). Comparison of Experiments. *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, 93–102.

Bradley, R. C. (2005). Basic Properties of Strong Mixing Conditions. A Survey and Some Open Questions. *Probability Surveys*, *2*. https://doi.org/10.1214/154957805100000104

DeGroot, M. H. (1970). *Optimal Statistical Decisions*. McGraw-Hill.

Dudík, M., Langford, J., & Li, L. (2011). Doubly Robust Policy Evaluation and Learning. *Proceedings of the 28th International Conference on Machine Learning (ICML)*, 1097–1104. https://doi.org/10.1145/3106426.3106468

Ferguson, T. S. (1967). *Mathematical Statistics: A Decision Theoretic Approach*. Academic Press. https://doi.org/10.1016/B978-0-12-257650-3.50001-3

Hall, P. (1992). *The Bootstrap and Edgeworth Expansion*. Springer. https://doi.org/10.1007/978-1-4612-4384-7

Howard, S. R., Ramdas, A., McAuliffe, J., & Sekhon, J. (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, *49*(2), 1055–1080.

Jiang, N., & Li, L. (2016). Doubly Robust Off-policy Value Evaluation for Reinforcement Learning. *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, 652–661.

Kallus, N., & Uehara, M. (2020). Double Reinforcement Learning for Efficient Off-Policy Evaluation in Markov Decision Processes. *Journal of Machine Learning Research*, *21*(167), 1–63. http://jmlr.org/papers/v21/19-827.html

Kallus, N., & Uehara, M. (2022). Efficiently Breaking the Curse of Horizon in Off-Policy Evaluation with Double Reinforcement Learning. *Operations Research*, *70*(6), 3282–3302. https://doi.org/10.1287/opre.2021.2249

Lai, T. L., & Robbins, H. (1985). Asymptotically efficient adaptive allocation rules. *Advances in Applied Mathematics*, *6*(1), 4–22. https://doi.org/10.1016/0196-8858(85)90002-8

Lattimore, T., & Szepesvári, C. (2020). *Bandit Algorithms*. Cambridge University Press. https://doi.org/10.1017/9781108571401

Liu, Q., Li, L., Tang, Z., & Zhou, D. (2018). Breaking the Curse of Horizon: Infinite-Horizon Off-Policy Estimation. *Advances in Neural Information Processing Systems (Neurips)*, *31*, 5356–5366. https://doi.org/10.48550/arXiv.1810.12429

Newey, W. K., & West, K. D. (1987). A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. *Econometrica*, *55*(3), 703. https://doi.org/10.2307/1913610

Precup, D., Sutton, R. S., & Singh, S. P. (2000). Eligibility Traces for Off-Policy Policy Evaluation. *Proceedings of the 17th International Conference on Machine Learning (ICML)*, 759–766.

Ramdas, A., Grünwald, P., Vovk, V., & Shafer, G. (2023). Game-Theoretic Statistics and Safe Anytime-Valid Inference. *Statistical Science*, *38*(4), 576–601. https://doi.org/10.1214/23-STS894

Saito, Y., Aihara, S., Matsutani, M., & Narita, Y. (2021,). Open Bandit Dataset and Pipeline: Towards Realistic and Reproducible Off-Policy Evaluation. *Advances in Neural Information Processing Systems 34, Datasets and Benchmarks Track*.

Shi, C., Zhang, S., Lu, W., & Song, R. (2022). Statistical Inference of the Value Function for Reinforcement Learning in Infinite Horizon Settings. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, *84*(3), 765–793. https://doi.org/10.1111/rssb.12465

Shi, C., Zhu, J., Shen, Y., Luo, S., Zhu, H., & Song, R. (2024). Off-Policy Confidence Interval Estimation with Confounded Markov Decision Process. *Journal of the American Statistical Association*, *119*(545), 273–284. https://doi.org/10.1080/01621459.2022.2110878

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press. https://doi.org/10.5555/3312046

Thomas, P. S., Theocharous, G., & Ghavamzadeh, M. (2015). High Confidence Off-Policy Evaluation. *Proceedings of the 29th AAAI Conference on Artificial Intelligence*, 3000–3006.

Torgersen, E. (1991). *Comparison of Statistical Experiments* (Vol. 36, Pt. 36, Chapter 36). Cambridge University Press.

van der Vaart, A. W. (1998). *Asymptotic Statistics*. Cambridge University Press. https://doi.org/10.1017/CBO9780511802256

Vovk, V., & Wang, R. (2021). E-values: Calibration, combination, and applications. *The Annals of Statistics*, *49*(3), 1736–1754.

Wald, A. (1950). *Statistical Decision Functions*. John Wiley & Sons. https://doi.org/10.1007/978-1-4612-0919-5

Waudby-Smith, I., Wu, L., Ramdas, A., Karampatziakis, N., & Mineiro, P. (2022,). *Anytime-valid off-policy inference for contextual bandits*.